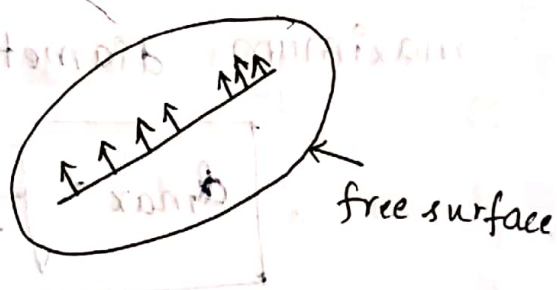
Surface Tension

* The property of free surface of the liquid acts as stretched elastic membrane & aquires minimum surface area is called surface tension.



* surface tension (T) $= \frac{F}{l}$

* units $\longrightarrow$ dyne/cm (C.G.S), N/m (S.I)

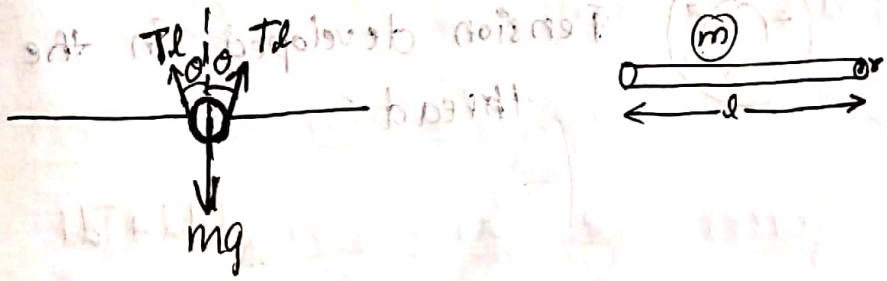
* It is a scalar quantity.

* Force due to surface tension,

$$\boxed{F = Tl}$$

Examples:

① Needle floating on the liquid surface:



since, needle is floating,

$$mg = 2Tl\cos\theta \quad (\because \theta \text{ is small}, \cos\theta \approx 1)$$

$$mg = 2T(l)$$

$$\pi r^2 \rho_{wire} l g = 2T l$$

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$$\pi r^2 \rho_{wire} g = 2T$$

maximum radius of wire,

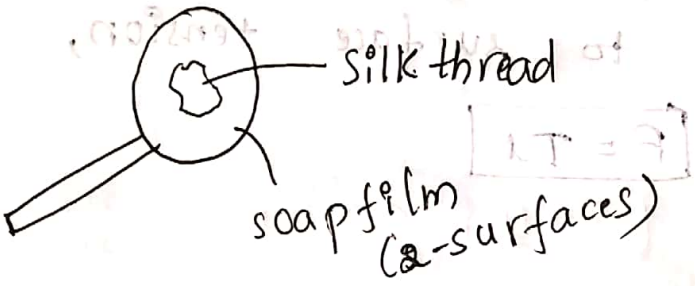
$$r_{max} = \sqrt{\frac{2T}{\pi \rho_{wire} g}}$$

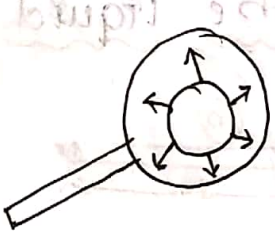
maximum diameter of wire,

$$d_{max} = \sqrt{\frac{2T}{\pi \rho_{wire} g}}$$

In this case, length of wire is "independent".

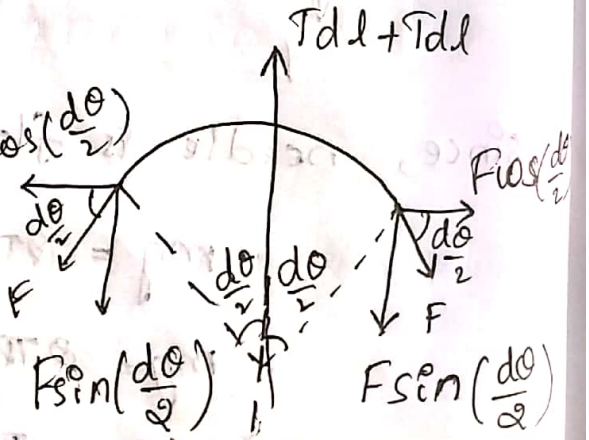
(2)





Tension developed in the thread.

In eqm,



$$2T dl = 2 \cdot F \sin\left(\frac{d\theta}{2}\right)$$

$$2T(R\,d\theta) = 2F \sin\left(\frac{d\theta}{2}\right)$$

$$2TR = F$$

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$F = 2\pi R$ $\quad L = 2\pi R$

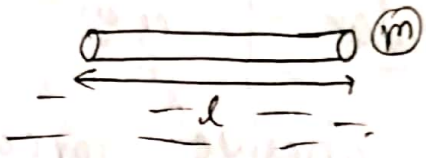
$R = \frac{L}{2\pi}$

$F = 2T \times \frac{L}{2\pi}$

$$\boxed{F = \frac{TL}{\pi}}$$ ✓

Force required to lift the floating objects from surface of liquid:

① For wire:

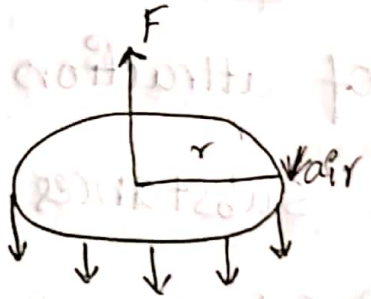


* $F = mg + T(2l)$

* extra force required (in addⁿ to wt)

$$\boxed{F = 2Tl}$$ ✓

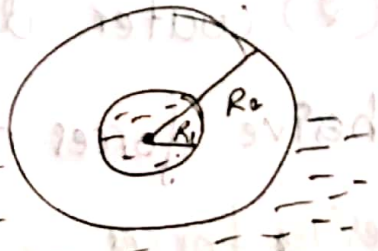
② For disc:



* $F = mg + 2\pi rT$

* $\boxed{F_{extra} = 2\pi rT}$ ✓

③ For concentric disc:



* $F = mg + T(2\pi R_1 + 2\pi R_2)$

* $\boxed{F_{extra} = T(2\pi R_1 + 2\pi R_2)}$ ✓

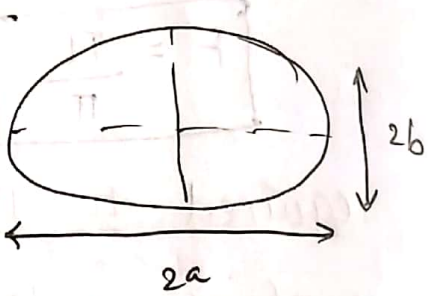
④ For Ring ($R_1 = R_2$):

* $F = mg + T(4\pi R)$

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* $\boxed{F_{extra} = T(4\pi R^2)}$

⑤ for elliptical plate



$F = mg + T\pi(a+b)$

* $\boxed{F_{extra} = T\pi(a+b)}$

Molecular forces :

(1) cohesive forces

(2) adhesive forces.

Cohesive forces :

Force of attraction b/w any 2 molecules of same substances.

Eg: (1) Glass molecules

(2) water drop

* Cohesive forces are strong in solids.

Adhesive forces :

Force of attraction b/w any 2 molecules of different substances.

* Best adhesive forces is b/w gum &

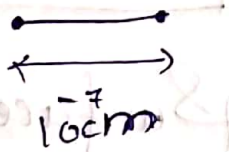
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papes.

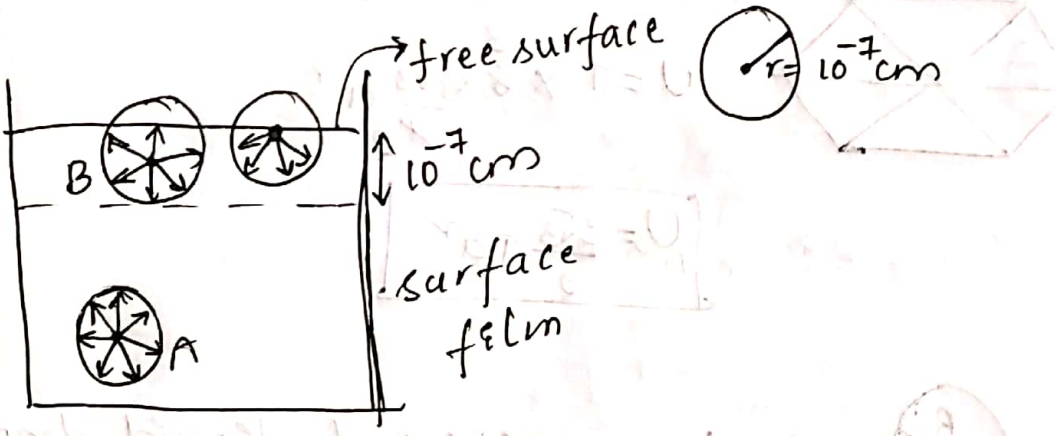
Molecular Range:

The max distance b/w 2 molecules upto which force of attraction exist b/w them

10^{-7} cm = 10^{-9} m



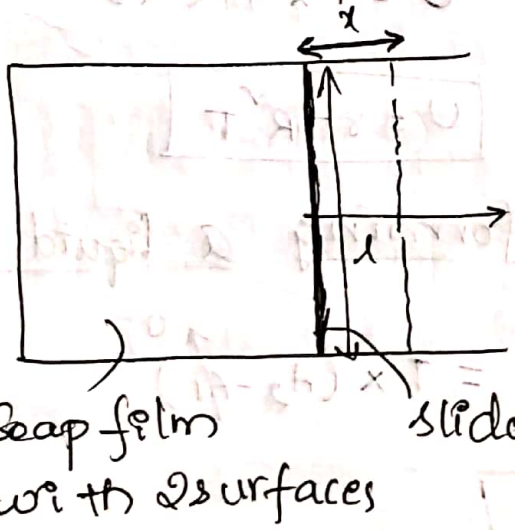
Sphere of Influence:



Work done = Surface energy

(depends on area of surface.

Relation b/w surface tension & surface energy:



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work done by Fx.

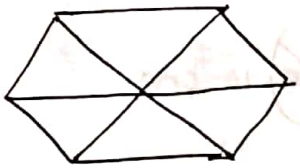
$$W = Fx$$

$$= T(2l)x$$

$$\boxed{W = T \times \Delta A}$$

∴ Surface energy $= T \times \Delta A$.

*



$$U = T \times 6 \times \frac{\sqrt{3}}{4} a^2$$

$$\boxed{U = \frac{3\sqrt{3}}{2} T a^2}$$

*



Surface energy of liquid drop

$$\boxed{U = T \times 4\pi R^2}$$

*



Surface energy of bubble (2-surfaces)

$$U = T \times 4\pi R^2 \times 2$$

$$\boxed{U = 8\pi R^2 T}$$

Work done in forming a liquid drop:

$$W = T \times \Delta A = T \times (A_2 - A_1)$$

($A_1 \to 0$)



$$\boxed{W = 4\pi R^2 T}$$

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Here ① $W \propto R^2 \Rightarrow \boxed{\frac{W_1}{W_2} = \left(\frac{R_1}{R_2}\right)^2}$

② $W \propto A$ ← surface area $\Rightarrow \boxed{\frac{W_1}{W_2} = \frac{A_1}{A_2}}$

③ volume of drop, $V = \frac{4}{3}\pi R^3$

$$R = \left(\frac{3V}{4\pi}\right)^{1/3}$$

∴ work done,

$$W = 4\pi R^2 T$$

$$W = 4\pi \left(\frac{3V}{4\pi}\right)^{2/3} T$$

$$W \propto V^{2/3}$$

$$\boxed{\frac{W_1}{W_2} = \left(\frac{V_1}{V_2}\right)^{2/3}}$$

Work done in blowing a soap bubble:

It has two surfaces



∴ $W = T \times \Delta A$

$$W = T\,(4\pi R^2)\,(2)$$

$$\boxed{W = 8\pi R^2 T}$$

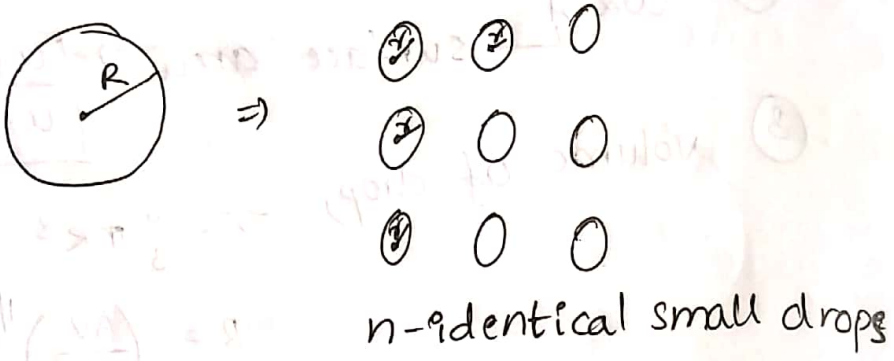
① $W \propto R^2$

② $W \propto A$

③ $W \propto (V)^{2/3}$

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Splitting of a big ~~bubble~~ drop into n-identical small drops :-



Here ① V is const

② $\frac{4}{3}\pi R^3 = n \times \frac{4}{3}\pi r^3$

$$R^3 = nr^3$$

$$\boxed{R = n^{1/3} r}$$

② Surface area increases.

$$\Delta A = A_2 - A_1$$

$$= n\,4\pi r^2 - 4\pi R^2$$

$$= 4\pi (nr^2 - R^2)$$

$$= 4\pi (nr^2 - n^{2/3} r^2)$$

$$\boxed{\Delta A = 4\pi r^2 (n - n^{2/3})}$$

(or)

$$\Delta A = 4\pi \frac{R^2}{n^{2/3}} (n - n^{2/3})$$

$$\boxed{\Delta A = 4\pi R^2 (n^{1/3} - 1)}$$

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(or) $V = \frac{4}{3}\pi R^3 \Rightarrow 4\pi R^2 = \frac{3V}{R}$

$$\Delta A = \frac{3V}{R}\left(\frac{R}{r} - 1\right)$$

$$\boxed{\Delta A = 3V\left(\frac{1}{r} - \frac{1}{R}\right)}$$

now, work done against the surface tension,

$$W = T \times \Delta A$$

$\Rightarrow$

$W = 4\pi r^2 T(n - n^{2/3})$
$W = 4\pi R^2 T(n^{1/3} - 1)$
$W = 3VT\left(\frac{1}{r} - \frac{1}{R}\right)$

**

* Since, surface area increases.

∴ Energy is ~~occupied~~ supplied to the system.

* If splitting occurs isothermally then energy is supplied to the system to keep temp of formed tiny droplets remains same.

* If splitting occurs adiabatically,

then $\boxed{dq = 0}$

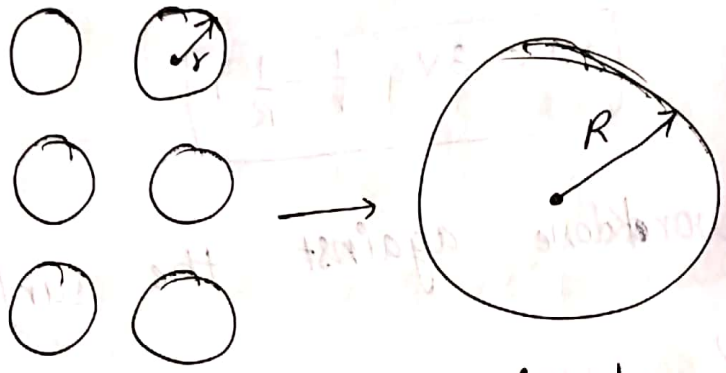
∴ system uses it's internal energy for splitting.

So temp of formed droplets decreases &

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hence becomes cool.

Merging of small droplets into bigger droplet:



n-small drops big drop.

here

① volume is const.

$$n\frac{4}{3}\pi r^3 = \frac{4}{3}\pi R^3$$

$$\boxed{n^{1/3} r = R}$$

② surface area decreases.

$$\Delta A = A_1 - A_2$$

$$= n 4\pi r^2 - 4\pi R^2$$

$$= 4\pi (n r^2 - R^2)$$

$$= 4\pi (n r^2 - n^{2/3} r^2)$$

$$= 4\pi r^2 (n - n^{2/3})$$

$$\boxed{\Delta A = 4\pi r^2 (n - n^{2/3})}$$

$$\boxed{\Delta A = 4\pi R^2 (n^{1/3} - 1)}$$

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$$V = \frac{4}{3}\pi R^3 \Rightarrow 4\pi R^2 = \frac{3V}{R}$$

$$\boxed{\Delta A = \frac{3V}{R}\left(\frac{R}{r} - 1\right)}$$

$$\boxed{\Delta A = \frac{3V}{R}\left(n^{1/3} - 1\right)} \quad \left(\because n^{1/3} = \frac{R}{r}\right)$$

now, workdone by the surface tension,

$$W = T(4\pi R^2)(n - n^{2/3})$$

$$W = T(4\pi R^2(n^{1/3} - 1)$$

$$W = T\left(\frac{3V}{R}\right)(n^{1/3} - 1)$$

* Since, surface area decreases.

  Energy is supplied by the system.

* If merging occurs isothermally, then energy supplied by the system to keep the temp of formed bigger drop remains same.

* If merging occurs adiabatically,

$$\boxed{dq = 0}$$

① Internal energy is utilized in K.E of big drop, i.e, $\boxed{W = K.E}$

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$$3\pi T\left(\frac{1}{r}-\frac{1}{R}\right)=\frac{1}{2}MV^2$$

$$3\pi T\left(\frac{1}{r}-\frac{1}{R}\right)=\frac{1}{2}\pi s V^2$$

$$V^2=\frac{6T}{s}\left(\frac{1}{r}-\frac{1}{R}\right)$$

$$\boxed{V=\sqrt{\frac{6T}{s}\left(\frac{1}{r}-\frac{1}{R}\right)}}$$

speed of bigger drop.

(2) Internal energy is utilized in raising the temp of bigger drop.

$$W=J\theta \quad (\text{Joule's law})$$

$$3\pi T\left(\frac{1}{r}-\frac{1}{R}\right)=Jmc\Delta\theta$$

$$3\pi T\left(\frac{1}{r}-\frac{1}{R}\right)=J\pi sc\Delta\theta$$

$$\boxed{\Delta\theta=\frac{3T}{Jsc}\left(\frac{1}{r}-\frac{1}{R}\right)}$$

m = mass of drop.

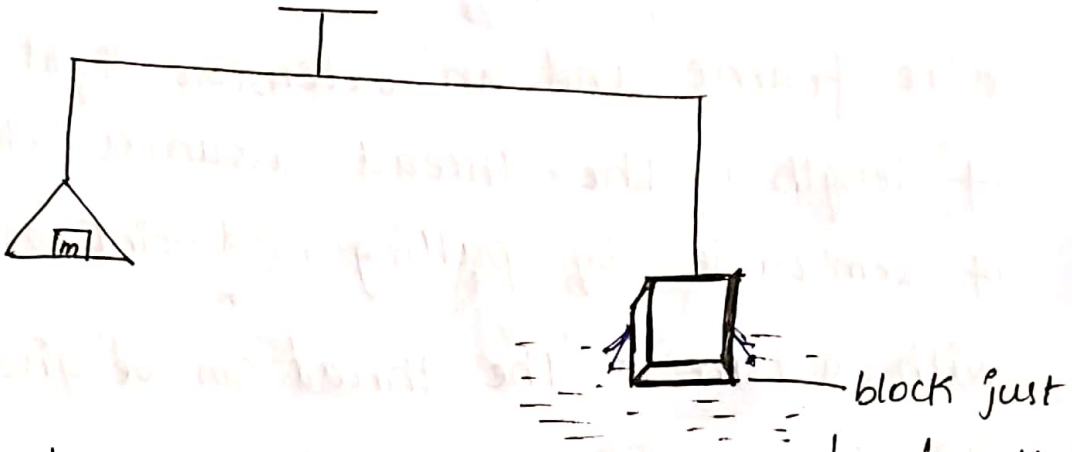
c = Specific heat of drop.

$\Delta\theta$ = raise in temp.

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Determination of surface tension of liquids

①



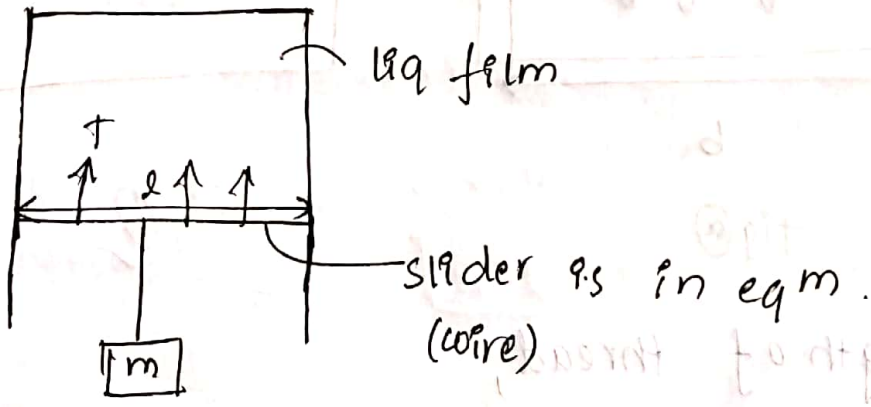
m → extra mass added in the pan in addn to wt of block.

* In eqm,

$$mg = 2(l+b)T$$

$$\Rightarrow \boxed{T = \frac{mg}{2(l+b)}}$$

②



* In eqm,

$$2Tl = mg$$

$$\boxed{T = \frac{mg}{2l}}$$

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Eg: when a film of soap soln is created inside a loop formed by rectangular wire frame and an extensible light thread of length L. The thread assumes shape of semicircle by pulling mid point of thread with a force F. The thread can be given shape of two semicircles as shown in fig. Find

(1) Surface Tension (T) of soap soln.

(2) Ratio of initial to final Tension in thread

(3) Heat generated in this process.

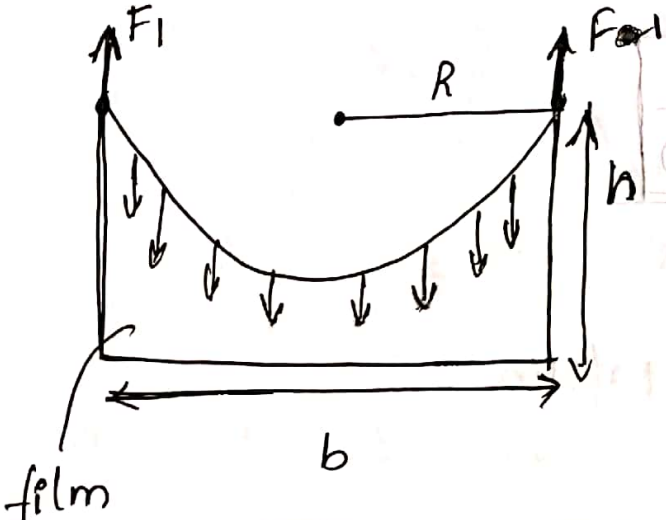


fig (a)

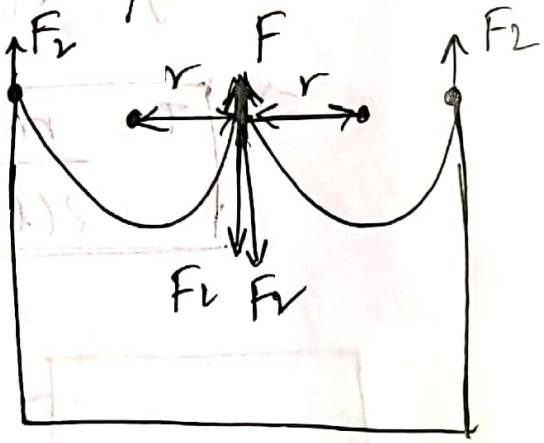


fig (b)

length of thread,

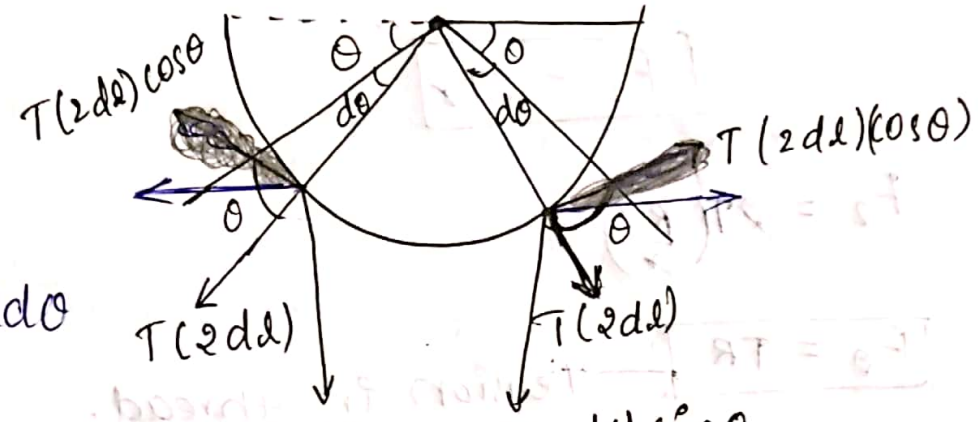
$$L = \pi R = \pi r + \pi r$$

i.e, $\boxed{R = 2r}$ — (1)

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* Force on thread due to surface tension

$$F_{ST} = \int_{\theta=0}^{\theta=\pi} 2T \cdot dl \sin\theta$$



Here,

$dl = R d\theta$

$$F_{ST} = \int_{\theta=0}^{\theta=\pi} 2T (R d\theta) \sin\theta = 2TR \int_{\theta=0}^{\pi} \sin\theta \cdot d\theta$$

$$F_{ST} = 4TR \left[-\cos\theta\right]_0^{\pi} \Rightarrow \boxed{F_{ST} = 4TR}$$

$F_{ST} = 2 \times T \times 2R$

$2R$ ← projection length (for any shape)

Since thread is in eqm;

$F_{net} = 0$

$2F_1 = 4TR$

$\boxed{F_1 = 2TR}$ ← Tension in thread.

From fig (b)

In eqm,

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$$2F_2 + F = 2T(2r) + 2T(2r)$$

$$\boxed{2F_2 + F = 8Tr}$$

$$F = 2F_2$$

$$4F_2 = 8Tr$$

$$\boxed{F_2 = 2Tr}$$

$$F_2 = 2T\left(\frac{R}{r}\right)$$

$$\boxed{F_2 = TR}$$ — Tension in thread.

now $\frac{F_1}{F_2} = \frac{2TR}{TR} = 2$

$$\boxed{\text{ratio} = 2:1}$$

And also

$$2F_2 + F = 8Tr$$

$$2F = 8T(r)$$

$$F = 4Tr$$

$$\boxed{F = 2TR}$$

$$F = 2T(4\pi)$$

$$\boxed{F = \frac{2LT}{\pi}}$$

$$\Rightarrow \boxed{T = \frac{\pi F}{2L}}$$

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Heat Calculations

$$U_1 + W = U_f + H$$

$$2T\left(bh - \frac{\pi R^2}{2}\right) + FR = 2T\left[bh - \frac{\pi r^2}{2} - \frac{\pi r^2}{2}\right] + H$$

$$2T\left(bh - \frac{\pi R^2}{2}\right) + FR = 2T_0\left(bh - \pi r^2\right) + H$$

$$2T\left(bh - \frac{\pi R^2}{2} - bh + \pi r^2\right) + FR = H$$

$$2T\left(\pi r^2 - \frac{\pi R^2}{2}\right) + F\frac{L}{\pi} = H$$

$$2T\left(\frac{\pi R^2}{4} - \frac{\pi R^2}{2}\right) + \frac{FL}{\pi} = H$$

$$T\left(\frac{\pi R^2}{2} - \pi R^2\right) + \frac{FL}{\pi} = H$$

$$H = \frac{FL}{\pi} - \pi\left(\frac{\pi R^2}{2}\right)$$

$$= \frac{FL}{\pi} - T\left(\frac{\pi \times L^2}{2 \times \pi^2}\right)$$

$$= \frac{FL}{\pi} - \frac{\pi F}{2L}\left(\frac{L^2}{2\pi}\right)$$

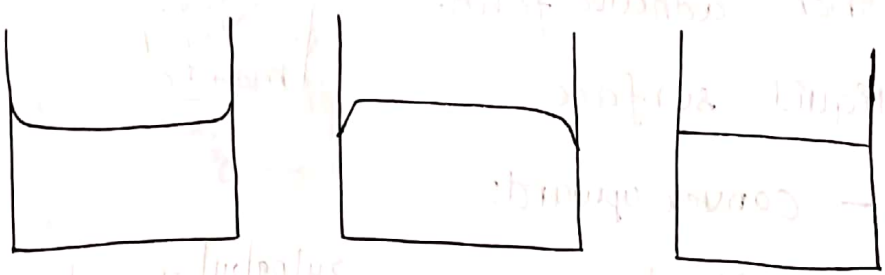
$$H = \frac{FL}{\pi} - \frac{FL}{4}$$

$$\boxed{H = FL\left(\frac{1}{\pi} - \frac{1}{4}\right)}$$

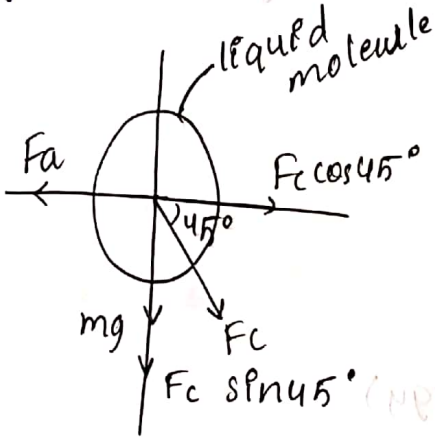
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Surface Tension (continuation)

Shape of free surface of liquid:
(meniscus)



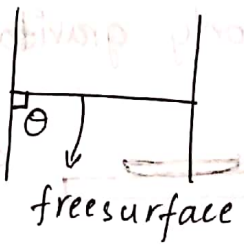
Here molecular forces decide the shape of liquid surface.



Possibilities:

① $F_a = F_c \cos 45° \Rightarrow F_a = \frac{F_c}{\sqrt{2}}$

then $F_{net} = mg + F_c \sin 45° (\downarrow)$

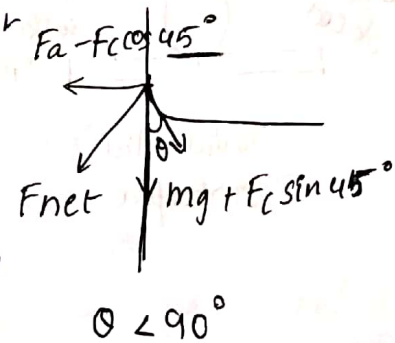


θ → angle of contact = 90°

② $F_a > F_c \cos 45° \Rightarrow F_a > \frac{F_c}{\sqrt{2}}$

i.e, Adhesive forces are stronger than cohesive forces.

shape of liquid surface
— concave upwards

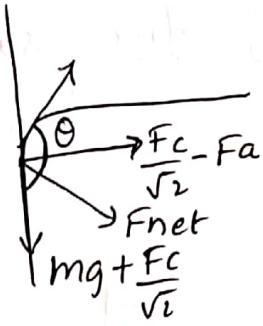


θ < 90°

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③ $F_a < F_c \cos 45^\circ \Rightarrow F_a < \frac{F_c}{\sqrt{2}}$

i.e, cohesive forces are stronger than adhesive forces.



shape of liquid surface.

– convex upwards

$\theta > 90^\circ$.

24|09|24| Tuesday

Shape of a liquid drop:

*Shape of a liquid drop is defined by

① Gravitational force

② Surface tension force

*In eqm,

P.E is minimum.

L (grav. P.E + surface energy)

1) If only surface tension forces are considered :-



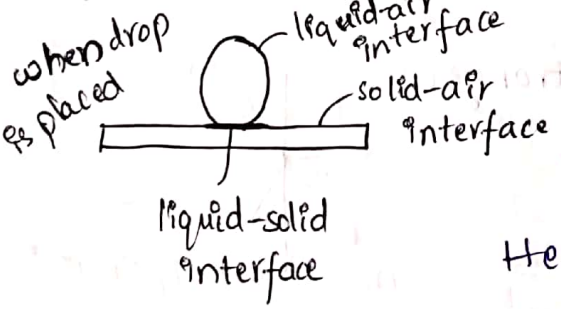
drop is perfect sphere.

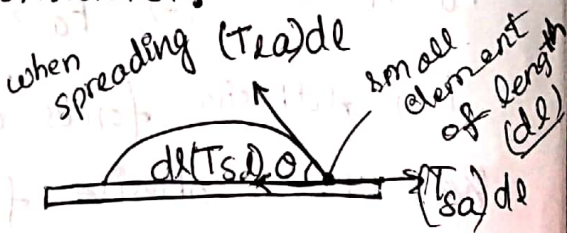
2) If only gravitational forces are considered :-



drop becomes flat (disc).

3) If both the forces are considered :-





Here,

Area of sa interface – ↓

Area of l-a interface – ↑

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Area of solid-liquid ~~surface~~ interface — ↑

$U_{sa} = T_{sa} \cdot A_{sa}$ (↓)

$U_{la} = T_{la} \cdot A_{la}$ (↑)

$U_{sl} = T_{sl} \cdot A_{sl}$ (↑)

} dominates

In eqm,

Net force on the element = 0

$$T_{sl} \cdot dl + T_{la} \cdot dl \cos\theta = T_{sa} \cdot dl$$

$$\boxed{\cos\theta = \frac{T_{sa} - T_{sl}}{T_{la}}}$$

Angle of Contact:

* It is the angle b/w tangent drawn to the liquid surface at the pt of contact & solid surface inside the liquid.

* θ lies in b/w $0°$ to $180°$.

Eg: For pure water & glass, $\theta = 0°$

For ordinary tap water, $\theta = 8°$

For mercury & glass, $\theta \approx 140°$

For pure water in silver vessel, $\theta = 90°$

* If $\theta < 90°$, adhesive forces dominates the cohesive forces. those liquids wet the solids.

* If $\theta > 90°$, cohesive forces dominates the adhesive

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forces, those liquids will not wet the solids.

* θ depends on nature of solid & liquid pair in contact.
* θ depends on medium above the liquid surface.
* θ depends on impurities added to the liquid.
* For partially soluble impurities will decrease θ.
  Eg: detergents, soaps.
* For completely soluble impurities will increase θ.
  Eg: NaCl (salt water)
* θ depends on temperature.
* As temp ↑, θ ↓.
* water proofing agents, increases θ.
  Eg: wax in rain coats, umbrellas, etc.
* Surface tension of a liquid also depends on

① Impurities added to it:-

a) partially soluble — S.T ↓

b) completely soluble — S.T ↑

② Temperature:-

we have,

$$S_t = S_0 (1 - \alpha t)$$

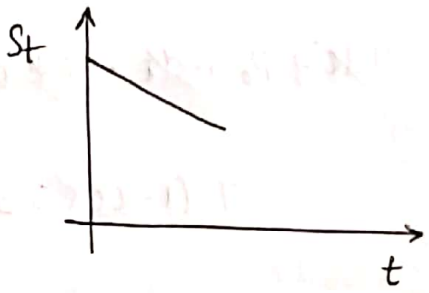
S_t = surface tension of liq at $t^0 C$

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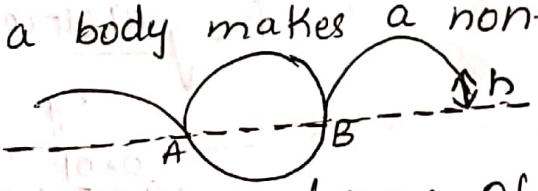
S_0 = surface tension of liquid at 0°c.

α = positive const.

t = temp of liquid.



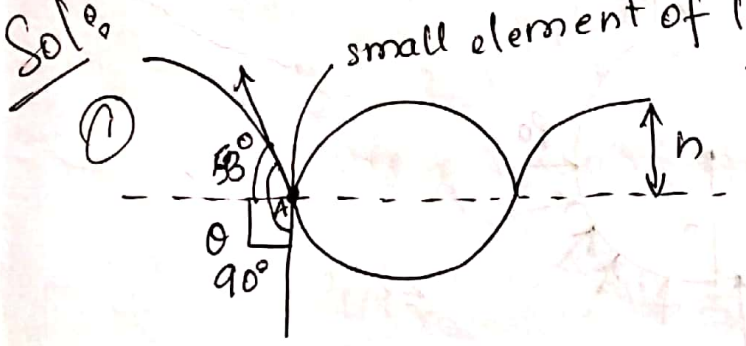
Eg: Line AB represents diameter of sphere. Assume liquid & material of a body makes a non-wetting comb^n with contact angle 143° at the circumference of sphere passing through AB. surface tension is 0.05N, ρ of liq = 4×10^3 kg/m^3, material of radius of sphere = 2 mm, $g=10\ m/s^2$. The extra polated line AB is at a depth 'h' below the horizontal flat part of free surface. Find

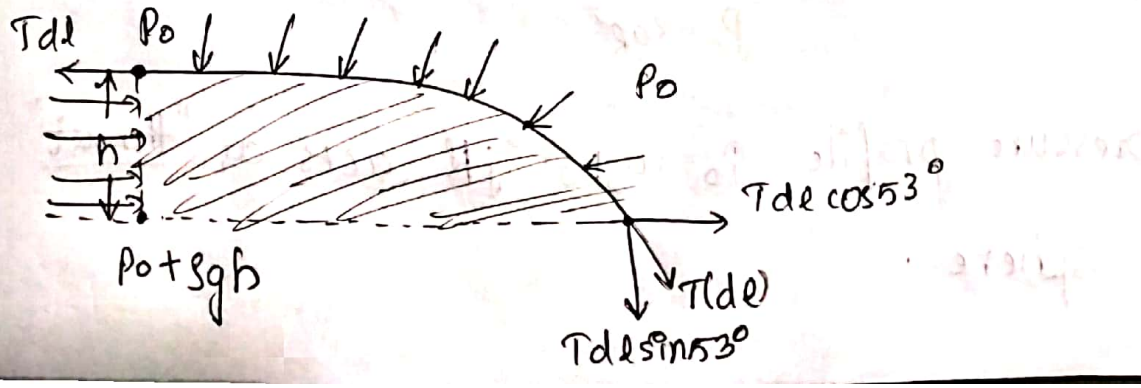


① 'h'

② density of material of sphere.

Sol^n





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In eqm,

$$T\,dl + P_0 h \cdot dl = T\,dl\cos 53^\circ + \left(\frac{2P_0 + \rho g h}{2}\right) h\,dl$$

$$T(1-\cos 53^\circ) = \frac{\rho g h^2}{2}$$

$$h = \sqrt{\frac{2T(1-\cos 53^\circ)}{\rho g}}$$

$$h = \sqrt{\frac{2 \times 0.05 \times 2/5}{4\times10^3 \times 10}}$$

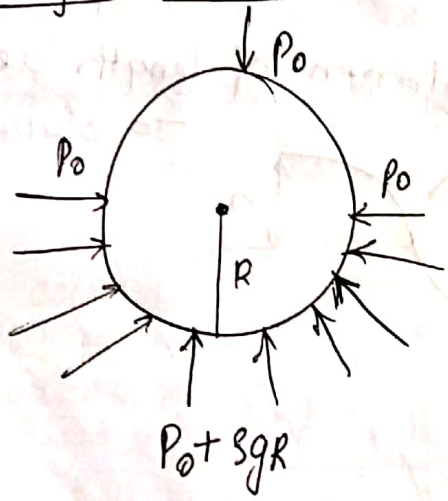
$$h = \sqrt{\frac{4\times10^{-2}}{4\times10^{4}}} = \sqrt{10^{-6}}$$

$$h = 10^{-3}\,m$$

$$h = 1\,mm$$

(2) to find density of material of sphere :

Without Surface Tension :



pressure profile P_0 to $P_0 + \rho g R$ acts as "thrust" on sphere.

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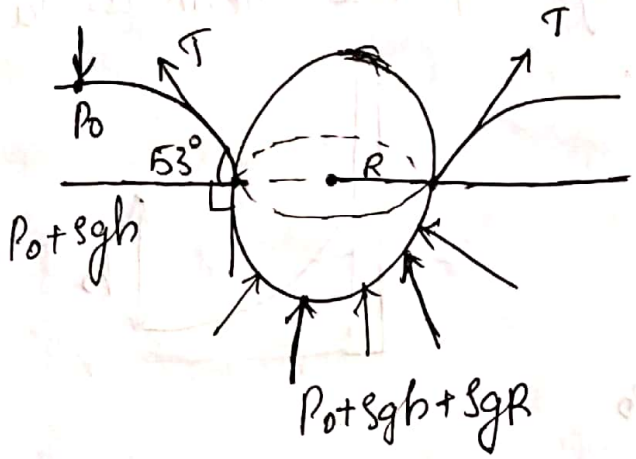
In eqm,

$$F_B = mg$$

$$\frac{2}{3}\pi R^3 s_l g = \frac{4}{3}\pi R^3 s_b g$$

$$\boxed{s_l = 2 s_b}$$

With Surface Tension:



$sgh \rightarrow$ uniform.

In eqm,

wt of sphere = thrust + $sgh(\pi R^2)$ + F_{ST}

$$\frac{4}{3}\pi R^3 s_b g = \frac{2}{3}\pi R^3 s g + sgh(\pi R^2) + T\sin 53^\circ (2\pi R)$$

$$\frac{4}{3} R^3 s_b g = \frac{2}{3} R^3 s g + sghR + T\sin 53^\circ (2)$$

$$\frac{4}{3} \times 4 \times 10^{-6} \times s_b \times 10 = \frac{2}{3} \times 4 \times 10^{-6} \times 10^3 \times 4 \times 10 + 4 \times 10^{-2} \times 10 \times 10^{-2} \times 2 \times 10^{-2}$$

$$+ 0.005 \times \frac{4}{5} \times 2$$

$$\frac{16}{3} \times 10^{-5} \times s_b = \frac{32}{3} \times 10^{-2} + 8 \times 10^{-2} + 8 \times 10^{-4}$$

$$\frac{16}{3} \times 10^{-3} \times s_b = \frac{32 + 24 + 24}{3}$$

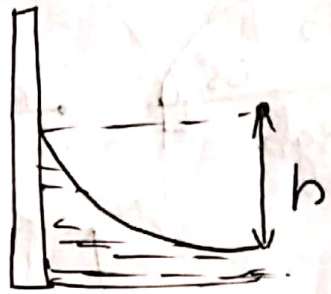
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$$2 \times 10^{-3} \times \rho_b = 4 + 3 + 3$$

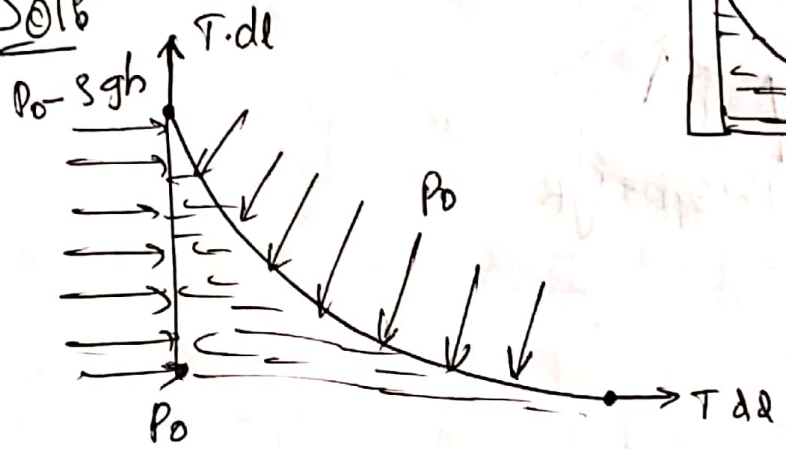
$$\rho_b = \frac{10}{2} \times 10^{+3}$$

$$\boxed{\rho_b = 5000 \text{ kg/m}^3}$$

Eg: water in a aquarium forms a meniscus as shown. Calculate the diff in height 'h' b/w centre & edge of the meniscus. Take T is 0.073 N/m.



Solⁿ



In eqm, net force on liquid column = 0

$$T\,dl + \left[\frac{P_0 + (P_0 - \rho g h)}{2}\right] h\,dl = P_0 h\,dl$$

$$0.073 + P_0 h - \frac{\rho g h^2}{2} = P_0 h$$

$$\frac{0.073 \times 2}{10^3 \times 10} = h^2$$

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$$h^2 = \frac{146}{10^7} = \frac{14.6}{10^6}$$

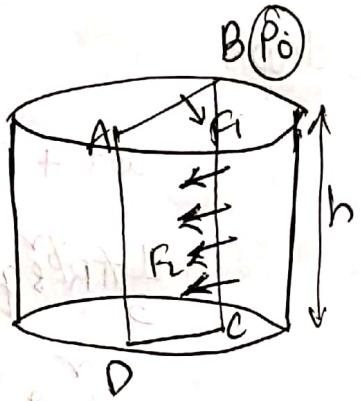
$$h = \frac{\sqrt{14.6}}{10^{-3}}$$

$$h = \sqrt{14.6} \times 10^{-3} m$$

$$\boxed{h \approx 3.8 mm}$$

Ex: Consider a vertical section ABCD of water column through a diameter of the beaker. Find the force on water on side of this section by water on the other side of this section.

Sol:



net force $= F_2 - F_1$

where $F_1 = T(2R)$

& $F_2 = \left[\frac{P_0 + P_0 + \rho g h}{2}\right] h (2R)$

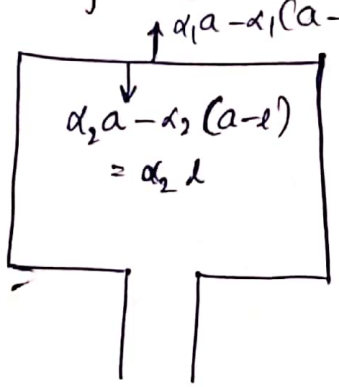
$$F_2 = (2P_0 + \rho g h) h R$$

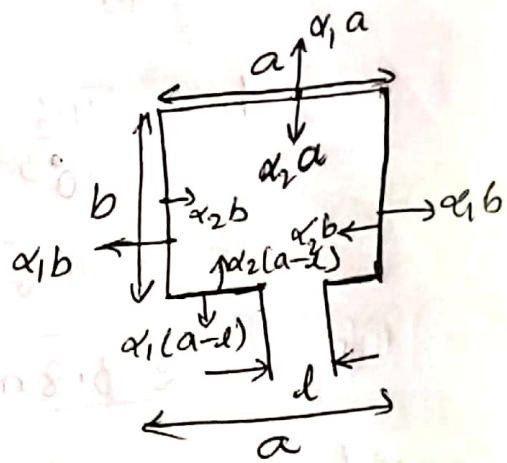
Ex: A light open rigid wire frame floats on the surface of water as shown. what force will act on the frame immediately after soap soln is dropped inside if α_1 & α_2 are S.Ts of water & soap. $(\alpha_1 > \alpha_2)$. what is the

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net force on this frame?

Sol:

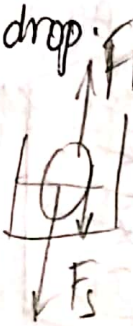




$$F_{net} = \alpha_1 l - \alpha_2 l$$

$$= (\alpha_1 - \alpha_2) l$$

Ex: Drops of liq of density s are floating ½ immersed in a liq of density σ. If surface Tension of liq is T. Find radius of the drop.



$$wt + F_{ST} = F_b$$

$$\frac{4}{3}\pi R^3 sg + T(2\pi R) = \frac{2}{3}\pi R^3 \sigma g$$

$$\frac{4}{3} R^3 sg + 2T = \frac{2}{3} R^3 \sigma g$$

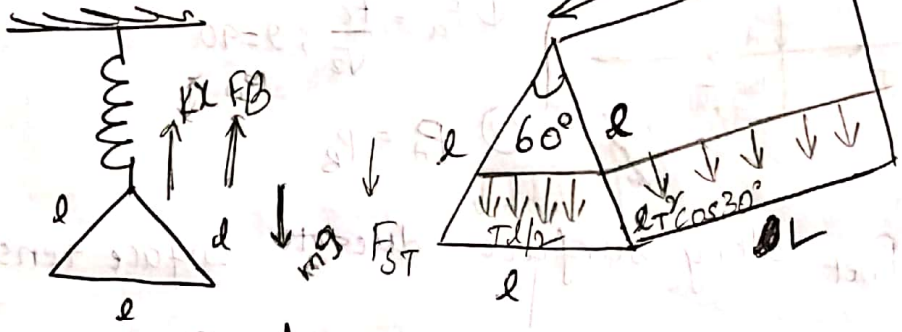
$$\frac{2}{3} R^3 (\sigma g - 2sg) = 2T$$

$$R^3 = \frac{3T}{\sigma g - 2sg}$$

$$\boxed{R = \sqrt{\frac{3T}{g(\sigma - 2s)}}}$$

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Eg: A glass prism of length l has its principle section of equilateral Δle of length l as shown. The prism with its base horizontal is supported by a vertical spring of force const (k) such that $1/2$ the slant surface of prism is submerged in water. T is surface Tension, $\theta = 0^\circ$, density of glass d, water s ($s < d$). Calculate the extension in the spring in this position of eqm.



cross-sectional view

Ans:

$$\frac{\sqrt{3}}{4}l^2 - \frac{\sqrt{3}}{4}\frac{l^2}{4} = \frac{\sqrt{3}}{4}l^2\left(1-\frac{1}{4}\right) = \frac{3\sqrt{3}l^2}{16}$$

$$x = \frac{1}{k}\left[\frac{\sqrt{3}}{4}l^2 L\,dg - \frac{3\sqrt{3}}{16}l^2 L s g + \sqrt{3}TL + Tl\right]$$

$$kx + F_B = mg + F_{ST}$$

$$kx + \frac{3\sqrt{3}l^2}{16}L s g = \frac{\sqrt{3}}{4}l^2 L\,dg + 2T L\cos 30^\circ + T\frac{l}{2}$$

$$kx = \frac{\sqrt{3}}{4}l^2 L dg + \sqrt{3}Tl + Tl - \frac{3\sqrt{3}}{16}l^2 L s g$$

$$\boxed{x = \frac{1}{k}\left[\frac{\sqrt{3}}{4}l^2 L dg + \sqrt{3}Tl + Tl - \frac{3\sqrt{3}}{16}l^2 L s g\right]}$$

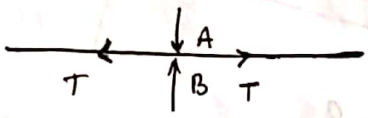
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N.T.Q

25/09/24 | wednesday

Pressure difference accross the liquid surface :

① Liquid surface is planar :

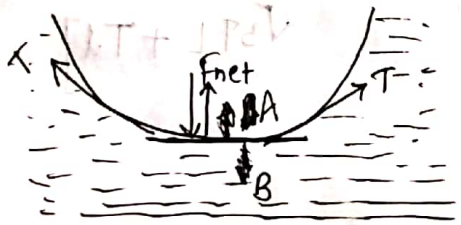


① $F_a = \frac{F_c}{\sqrt{2}}$; $\theta = 90°$

② $P_A = P_B$

Since F_{net} along surface due to surface tension is zero.

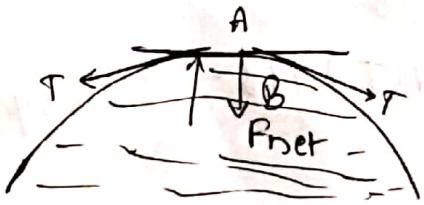
② For concave shape :-



① $F_a > \frac{F_c}{\sqrt{2}}$, $\theta < 90°$.

② $P_A > P_B$.

③ For convex shape :-



① $F_a < \frac{F_c}{\sqrt{2}}$, $\theta > 90°$

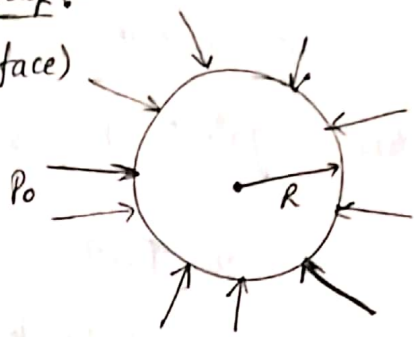
② $P_A < P_B$

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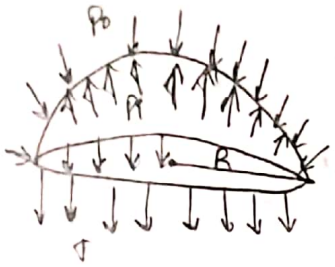
cess

pressure inside the liquid drop:-

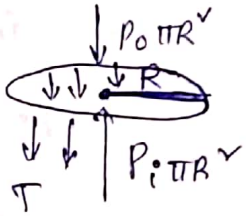
(single surface)



Method-1: Force Method:-



$\Rightarrow$



In eqm,

$$P_o \pi R^2 + 2\pi RT = P_i \pi R^2$$

$$P_o + \frac{2T}{R} = P_i$$

$$\Rightarrow \boxed{P_i = P_o + \frac{2T}{R}}$$

P_i — inside pressure

$\Rightarrow$ excess pressure $\Rightarrow P = P_i - P_o = \frac{2T}{R}$

$$\boxed{P_{excess} = \frac{2T}{R}}$$

Method-2: Energy Method:-



Initial surface Area

$$A_i = 4\pi R^2$$

Final Surface Area

$$A_f = 4\pi (R+dR)^2$$

Increase in surface area,

$$dA = A_f - A_i = 4\pi (R^2 + 2dR \cdot R + (dR)^2 - R^2)$$

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$$dA = 8\pi R dR$$

work done against the surface tension,

$$W = T \cdot dA$$

$$W = T(8\pi R \cdot dR) \quad -①$$

excess pressure at the inner surface,

$$P = P_i - P_o$$

work done by excess pressure,

$$W = P \times \text{increase in volume.}$$

$$W = (P_i - P_o) \times 4\pi R^2 dR \quad -②$$

From eqns ① & ②,

$$T(8\pi R \cdot dR) = (P_i - P_o) \times 4\pi R^2 \cdot dR$$

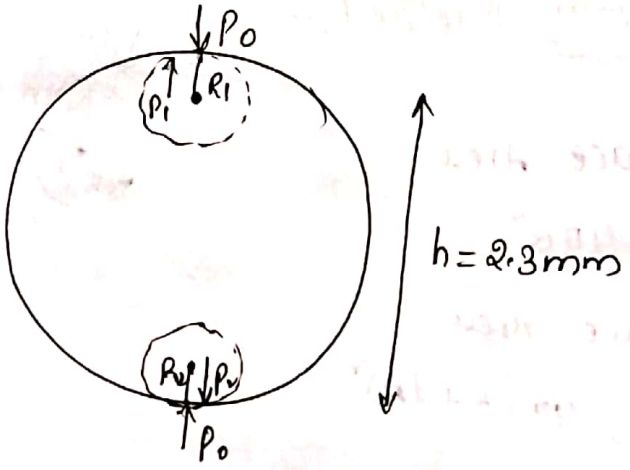
$$P_i - P_o = \frac{2T}{R}$$

$$\boxed{P_i = P_o + \frac{2T}{R}}$$

Eg: A rain drop is falling in air with a uniform velocity. Find the difference b/w curvature radius of drops surface at the upper and lower points of the drop seperated by a distance $h = 2.3mm$.

Sol:

For not a spherical drop:



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we have,

$$P_1 - P_0 = \frac{2T}{R_1} \quad -①$$

& $$P_2 - P_0 = \frac{2T}{R_2} \quad -②$$

now, $$P_2 - P_1 = 2T\left(\frac{1}{R_2} - \frac{1}{R_1}\right)$$

$$P_1 - P_2 = 2T\left(\frac{R_1 - R_2}{R_1 R_2}\right) \quad -③$$

Inside the drop,

$$P_2 = P_1 + \rho g h$$

$$P_2 - P_1 = \rho g h \quad -④$$

From eqn ③ & ④

$$\rho g h = \frac{2T(R_1 - R_2)}{R_1 R_2}$$

$$R_1 - R_2 = \frac{\rho g h (R_1 R_2)}{2T} \quad -⑤$$

Assume, $R_1 \approx R_2$ (approximately spherical drop)

$$R_1 + R_2 = h$$

$$2R_1 = h$$

$$R_2 = R_1 = h/2$$

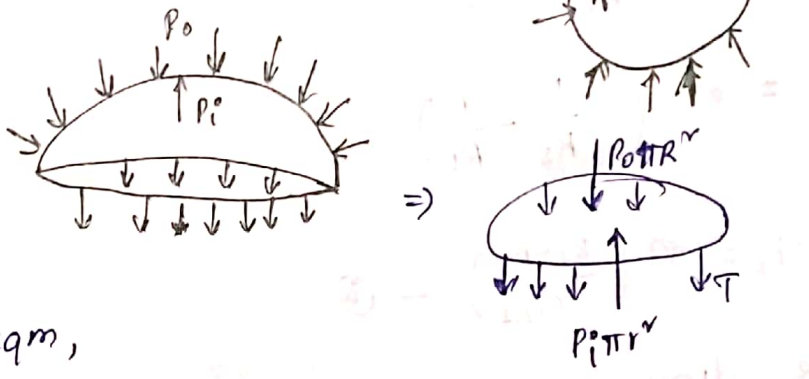
$$R_1 - R_2 = \frac{\rho g h}{2T}\left(\frac{h}{2}\right)\left(\frac{h}{2}\right)$$

$$\boxed{R_1 - R_2 = \frac{\rho g h^3}{8T}} \quad ***$$

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Excess pressure inside the soap bubble &
(for free surfaces)

Method-1: Force Method :-



In eqm,

$$P_o \pi R^2 + T(2\pi R)(2) = P_i \pi R^2$$

$$P_o + \frac{4T}{R} = P_i$$

$$\Rightarrow \boxed{P_i = P_o + \frac{4T}{R}}$$

$\Rightarrow$ excess pressure $\Rightarrow P = P_i - P_o$

$$\boxed{P_{excess} = \frac{4T}{R}}$$

Method-2: Energy Method :-

Initial Surface Area

$$A_i = 4\pi R^2 \times 2$$

Final surface Area

$$A_f = 4\pi (R+dR)^2 \times 2$$

Increase in surface area

$$dA = A_f - A_i$$

$$= 8\pi (R^2 + 2dR \cdot R + (dR)^2 - R^2)$$

$$dA = 16\pi R \cdot dR$$

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work done against the surface tension,

$$W = T \cdot dA$$

$$W = T(16\pi R\,dR) \quad -(1)$$

excess pressure at the inner surface,

$$P = P_i - P_o$$

work done by excess pressure,

$$W = P \times \text{increase in vol}$$

$$W = (P_i - P_o)\, 4\pi R^2 \cdot dR \quad -(2)$$

from eqns (1) & (2)

$$T(16\pi R \cdot dR) = P_i - P_o\,(4\pi R^2 \cdot dR)$$

$$\frac{4T}{R} = P_i - P_o$$

$$\boxed{P_i = P_o + \frac{4T}{R}}$$

Important Points :-

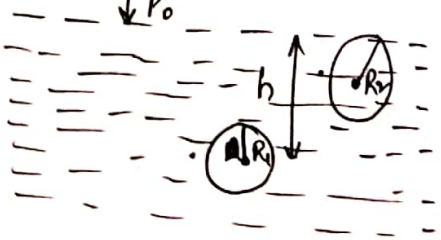
(1) Excess pressure in a liquid drop.

$$\boxed{P = \frac{2T}{R}} \Rightarrow \boxed{P \propto 1/R \;\&\; \frac{1}{\sqrt{A}} \;\&\; \frac{1}{\sqrt[3]{V}}}$$

↓

Valid even for bubble.

(2) Air bubble in liquid. (It has single surface)



$$\boxed{P_i = P_o + \rho g h + \frac{2T}{R_1}} \quad -(1)$$

when it is near to surface,

$h \approx 0$

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$$P_2 = P_0 + \frac{2T}{R_2} \quad -(2)$$

Temp of liq is const.

$$P_1V_1 = P_2V_2$$

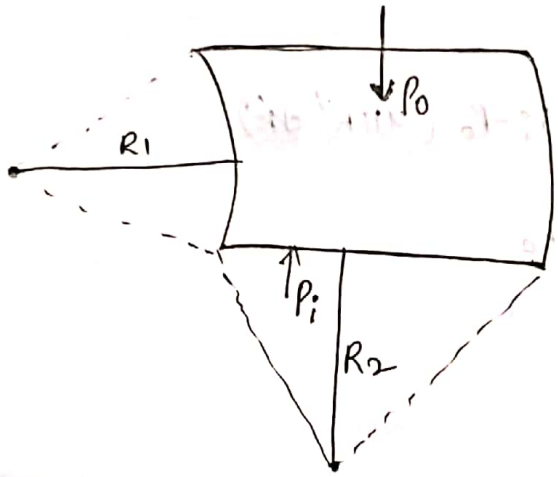
Here,

$P_0 = P_{atm} = 76\text{ cm of Hg} = 10\text{m of } H_2O = 1.013 \times 10^5\text{ Pa}$

$$P = hsg = \frac{76}{100} \times 13.6 \times 10^3 \times 9.8$$

$$P = Hsg = 10 \times 10^3 \times 9.8$$

(3) If the liquid surface has curvatures



$$P_i - P_0 = T\left(\frac{1}{R_1} + \frac{1}{R_2}\right)$$

* for liq film,

$$P_i - P_0 = 2T\left(\frac{1}{R_1} + \frac{1}{R_2}\right)$$

pts:

* for a spherical surface:

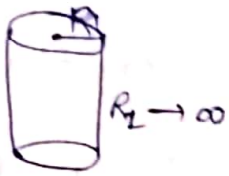
$$R_1 = R_2$$

$$P_i = P_0 + \frac{2T}{R}$$

* for a cylindrical surface:

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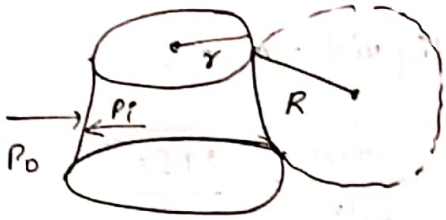
$$P_i - P_o = T\left(\frac{1}{R_1} + \frac{1}{R_2 \to \infty}\right)$$



$$\boxed{P_i - P_o = \frac{T}{R_1}}$$

★ Force required to seperate the glass plates when a liquid drop is trapped or placed b/w them.

★★ shape of drop b/w the plates.



$$P_o - P_i = T\left(\frac{1}{r} + \frac{1}{R}\right)$$

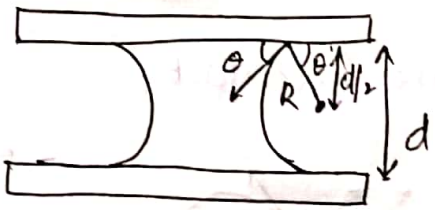
here $r >>> R$

$$\Rightarrow \frac{1}{R} >>> \frac{1}{r}$$

$$\therefore P_o - P_i = \frac{T}{R} \quad -(1)$$

d = distance b/w the plates.

$$\cos\theta = \frac{d}{2R}$$

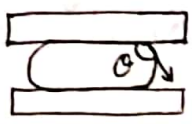


$$P_o - P_i = \frac{2T}{d}\cos\theta$$

for water drop $\Rightarrow \theta \to$ small

$$\cos\theta \approx 1$$

$$\boxed{P_o - P_i = \frac{2T}{d}}$$

for mercury drop $\Rightarrow$ 

★ A is area of plate which is in contact with film

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* force on the plates

$$F = (P_0 - P_i) A$$

$$\boxed{F = \frac{2TA}{d}}$$

* V = volume of liquid = Ad

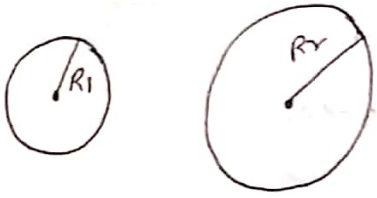
then $\boxed{F = \frac{2TV}{d^2} = \frac{2TA^2}{V}}$

* mass of liquid = m = ρV

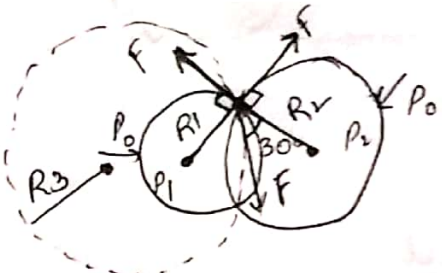
$$\boxed{F = \frac{2Tm}{\rho d^2} = \frac{2T\rho A^2}{m}}$$

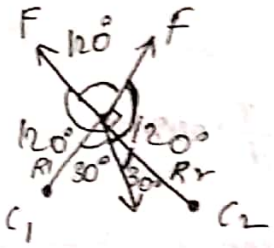
Merging of Soap bubbles

26|09|24| Thursday



when they are in contact,





From cosine law,

$$\cos 60^\circ = \frac{R_1^2 + R_2^2 - L^2}{2R_1R_2}$$

$$\frac{1}{2} \times 2R_1R_2 = R_1^2 + R_2^2 - L^2$$

$$L^2 = R_1^2 + R_2^2 - R_1R_2$$

$$\boxed{L = \sqrt{R_1^2 + R_2^2 - R_1R_2}}$$ → distance b/w centres of bubbles.

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To find radius of curvature of interface:

$$P_1 - P_0 = \frac{4T}{R_1}$$

& $$P_2 - P_0 = \frac{4T}{R_2}$$

$$P_1 - P_2 = 4T\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

since $R_1 < R_2 \Rightarrow \frac{1}{R_1} > \frac{1}{R_2}$

$$\boxed{P_1 > P_2}$$

$$P_1 - P_2 = \frac{4T}{R_3}$$

$$\frac{4T}{R_3} = 4T\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

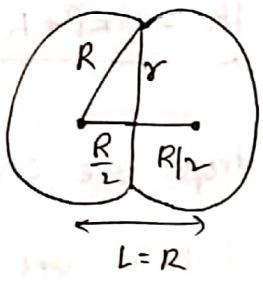
$$\frac{1}{R_3} = \frac{1}{R_1} - \frac{1}{R_2}$$

$$\boxed{R_3 = \frac{R_1 R_2}{R_2 - R_1}}$$

* If $R_1 = R_2 = R$ (say)

then $L = \sqrt{R^2 + R^2 - R^2} = R$

$$R_3 = \frac{R^2}{R - R} \Rightarrow \boxed{R_3 = \infty}$$



$$r = \sqrt{R^2 - \left(\frac{R}{2}\right)^2}$$

$$\boxed{r = \frac{\sqrt{3}R}{2}}$$

radius of circle of interface.

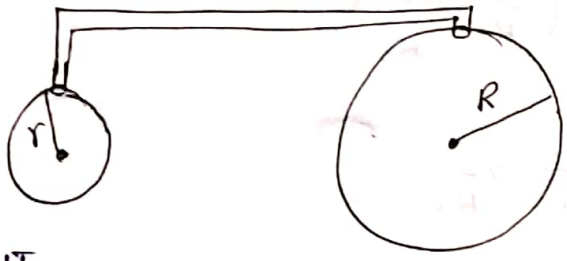
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∴ Area of circle,

$$A = \pi r^2 = \pi \left(\frac{\sqrt{3}R}{2}\right)^2 = \frac{3\pi R^2}{4}.$$

→ Coalesce of the bubbles:

combined / merging



$$P = \frac{4T}{r}$$

$r \downarrow \Rightarrow P \uparrow$

Cases:

① If the bubbles are coalesce isothermally in vaccum:

Total no. of moles = const

$$n = n_1 + n_2$$

$$PV = nRT \Rightarrow n = \frac{PV}{RT}$$

$$P_1V_1 + P_2V_2 = PV$$

$$\frac{4T}{R} \times \frac{4}{3}\pi R^3 = \frac{4T}{RT} \times \frac{4}{3}\pi R_1^3 + \frac{4}{3}\pi R_2^3 \times \frac{4T}{R_2}$$

$$R^2 = R_1^2 + R_2^2$$

$$\boxed{R = \sqrt{R_1^2 + R_2^2}}$$

* If two drops are coalsce $\Rightarrow V_1 = V_2$.

② If the bubbles are coalesce isothermally in atmosphere:

Total no. of moles = const

$$n = n_1 + n_2$$

$$PV = nRT \Rightarrow n = \frac{PV}{RT}$$

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$$PV = P_1V_1 + P_2V_2$$

$$\left(P_0 + \frac{4T}{R}\right) \times \frac{4}{3}\pi R^3 = \left(P_0 + \frac{4T}{R_1}\right)\frac{4}{3}\pi R_1^3 + \left(P_0 + \frac{4T}{R_2}\right)\cdot \frac{4}{3}\pi R_2^3$$

$$\left(\frac{P_0R + 4T}{R}\right)R^3 = \left(\frac{P_0R_1 + 4T}{R_1}\right)R_1^3 + \left(\frac{P_0R_2 + 4T}{R_2}\right)R_2^3$$

$$P_0R^3 + 4TR^2 = P_0R_1^3 + 4TR_1^2 + P_0R_2^3 + 4TR_2^2$$

$$\boxed{\frac{P_0(R^3 - R_1^3 - R_2^3)}{4(R_1^2 + R_2^2 - R^2)} = T}$$ – surface tension of liquid.

③ Again,

$$n = n_1 + n_2$$

$$\left(P_0 + \frac{4T}{R}\right)\frac{4}{3}\pi R_1^3 = \left(P_0 + \frac{4T}{R_1}\right)\frac{4}{3}\pi R_1^3 + \left(P_0 + \frac{4T}{R_2}\right)\frac{4}{3}\pi R_2^3$$

$$P_0\left(\frac{4}{3}\pi R^3 - \frac{4}{3}\pi R_1^3 - \frac{4}{3}\pi R_2^3\right) + \frac{4T}{3}\left(\frac{4\pi R^3}{R} - \frac{4\pi R_1^3}{R_1} - \frac{4\pi R_2^3}{R_2}\right)$$

$$= 0$$

$$P_0(\Delta V) + \frac{4T}{3}(\Delta A) = 0$$

$$\Rightarrow \boxed{3P_0(\Delta V) + 4T(\Delta A) = 0}$$

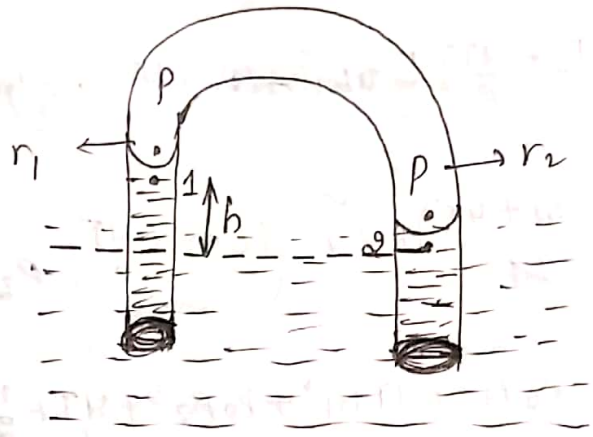
(ΔV: change in volume; ΔA: change in surface area.)

~~Manometer Including Surface Tension:~~

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Manometer Including Surface tensions

*



$$P_0 - \frac{2T}{r_1} = P_1 \quad -(1)$$

$$P - \frac{2T}{r_2} = P_2 \quad -(2)$$

$$P_1 - P_2 = 2T\left(\frac{1}{r_2} - \frac{1}{r_1}\right)$$

$$P_2 - P_1 = 2T\left(\frac{1}{r_1} - \frac{1}{r_2}\right)$$

Again,

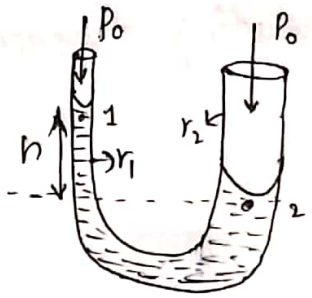
$$P_1 + sgh = P_2$$

$$P_2 - P_1 = sgh$$

$$sgh = 2T\left(\frac{1}{r_1} - \frac{1}{r_2}\right)$$

$$\boxed{h = \frac{2T}{sg}\left(\frac{1}{r_1} - \frac{1}{r_2}\right)}$$

*



$$P_0 - \frac{2T}{r_1} = P_1 \quad -(1)$$

$$P_0 - \frac{2T}{r_2} = P_2 \quad -(2)$$

$$P_1 + sgh = P_2 \quad -(3)$$

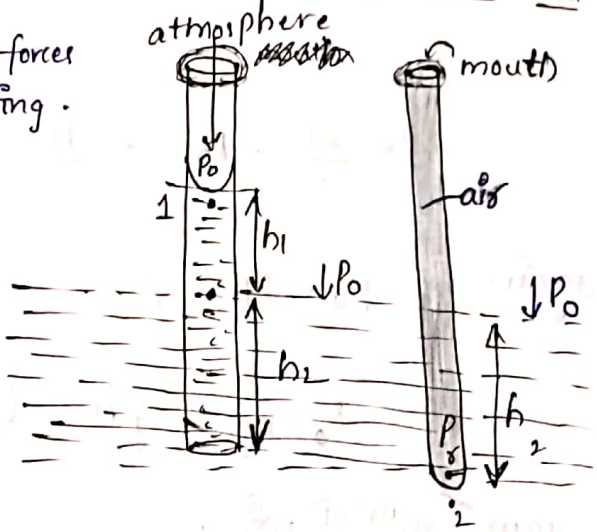
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From eqn ①, ②, ③

$$\boxed{h = \frac{2T}{\rho g}\left(\frac{1}{r_1} - \frac{1}{r_2}\right)}$$

Pressure required to blow a soap bubble from lower end of tube is

adhesive forces dominating.



At Pt-2

$$P - P_2 = \frac{2T}{r}$$

$$P = P_2 + \frac{2T}{r} \quad -①$$

and $$P_0 + \rho g h_2 = P_2 \quad -②$$

$$\therefore P = P_0 + \rho g h_2 + \frac{2T}{r} \quad -ⓐ$$

At Pt-1

$$P_0 - P_1 = \frac{2T}{r} \quad -①$$

$$P_1 + \rho g h_1 = P_0$$

$$\therefore P_0 - P_1 = \rho g h_1 \quad -②$$

$$\frac{2T}{r} = \rho g h_1$$

$$\boxed{h_1 = \frac{2T}{\rho g r}}$$

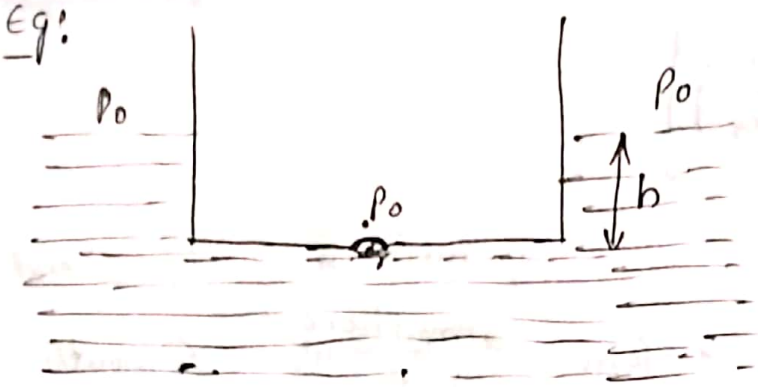
from ⓐ, $$P = P_0 + \rho g h_1 + \rho g h_2$$

$$P = P_0 + \rho g (h_1 + h_2)$$

$$P - P_0 = \boxed{P_{excess} = \rho g (h_1 + h_2)}$$

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Eg:



$$P - \frac{2T}{r} = P_0 \quad -(1)$$

again, $P_0 + \rho g h = P$

$$P_0 - P_0 = \rho g h \quad -(2)$$

from eqns (1), (2).

$$\rho g h = \frac{2T}{r}$$

$$\boxed{h = \frac{2T}{r \rho g}}$$

Eg: A mercury drop shaped as a round table of radius R & thickness h is located b/w 2 horizontal glass plates Assuming that $h \ll R$, Find the mass m of a wt which has to be placed on the copper plate to diminish the distance b/w the plates (n) times.

Ans: $m = \frac{2\pi R^2 T \cos\theta (n^2 - 1)}{gh}$

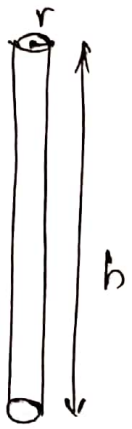


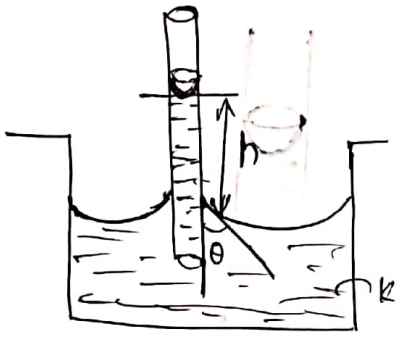
Sol:

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Capillary Tube:

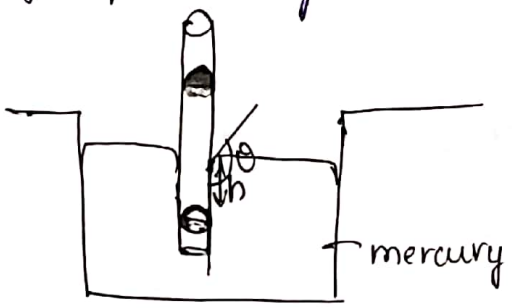
* A long glass tube having narrow bole is called cappilary tube. i.e, $r << h$.





rise of liquid is observed in kerosene.

** when tube is placed some what bent, then length of liquid change but angle of contact doesnot change. (θ)



fall of liquid is observed in mercury.

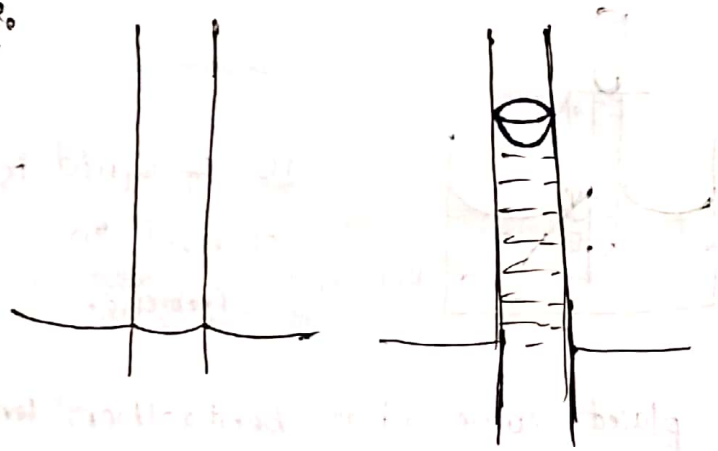
Capilarity:

* rise or fall of a liquid in capillary tube is called as capillarity.

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Ex: A sufficient long thin glass capillary tube of radius r is inserted vertically and held at rest into water. A const height of water column exist in the tube. Now another solid glass cylindrical rod of radius R/2 is inserted symmetrically into the glass capillary. water level in the capillary changes and attains a new fixed value. In the process total change in surface energy of glass & water combination on introducing rod in capillary is $\frac{x\pi T^2 r^3}{\rho g}$. Find x. (angle of contact = 0° & T, 2T represent surface tension of water-air interface, glass-air interface, ρ is density of water, g is acceleration due to gravity at the place, neglect water in the meniscus).

Sol:



A_{la} ⟶ increases

A_{sl} ⟶ increases

A_{sa} ⟶ decreases

we have,

Surface energy

$U_{la} = T_{la} \cdot A_{la}$ ⟶ increases

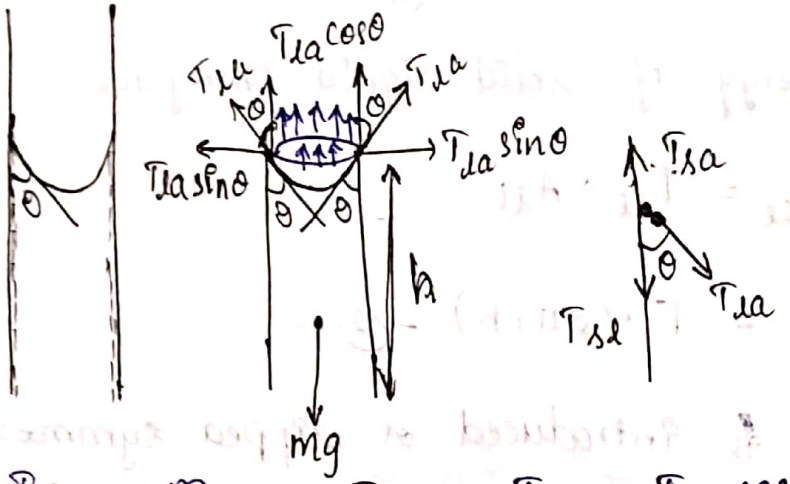
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$U_{sa} = T_{sa} \times A_{sa} \longrightarrow$ decreases

$\therefore$ dominates increase value

Becoz of ① gravitational P.E stored in liquid.

② Heat produced



In eqm, $T_{sa} = T_{sl} + T_{la}\cos\theta$

$$T_{sa} - T_{sl} = T_{la}\cos\theta$$

$$\frac{T_{sa} - T_{sl}}{\cos\theta} = T_{la}$$

$$\boxed{T_{la} = (T_{sa} - T_{sl})\sec\theta}$$

here $\theta = 0°$, $T_{sa} = 2T$, $T_{la} = T$

$$T_{sl} = 2T - T = T$$

$$\boxed{T_{sl} = T}$$

Rise of liquid in capillary tube.

wt balance method:

liquid column is in eqm.

$$T_{la}\cos\theta\,(2\pi r) = mg$$

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$$2\pi r \cdot T_{la}\cos\theta = \pi r^2 h \rho g$$

$$\boxed{h = \frac{2T_{la}\cos\theta}{\rho g r}}$$

we get, $\boxed{h = \frac{2T\cos\theta}{\rho g r}}$ — (1) $\quad T = T_{la}$

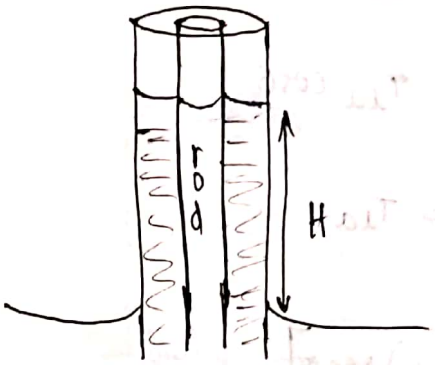
now,

surface energy of solid liquid interface.

$$U_{sl} = T_{sl} \cdot A_{sl}$$

$$= T \cdot (2\pi r h) \quad — (2)$$

now solid rod is introduced or dipped symmetrically



Again,

In eqm,

$$\left(2\pi r + 2\pi \frac{r}{2}\right) T_{la}\cos\theta = \left[\pi r^2 - \pi\left(\frac{r}{2}\right)^2\right] H \rho g$$

$$2\pi r T_{la}\cos\theta = \frac{3\pi r^2}{4} H \rho g$$

$$\boxed{\frac{4 T_{la}\cos\theta}{\rho g r} = H}$$

$\Rightarrow$ $\boxed{H = 2h}$

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now,

new surface energy of solid liquid interface.

$$U'_{sl} = T_{sl} \cdot A'_{sl}$$

$$= T\left(2\pi r H + 2\pi \frac{r}{2} \cdot H\right)$$

$$= 3\pi r H T$$

$$U'_{sl} = 6\pi r T h$$

Increase in surface energy of solid-liquid interface

$$\Delta U = U'_{sl} - U_{sl} = 4\pi r h T$$

$$\Delta U = 4\pi r^2 \left(\frac{2T\cos 0^\circ}{8gr}\right) T$$

$$\boxed{\Delta U = \frac{8\pi T^2}{8g}}$$

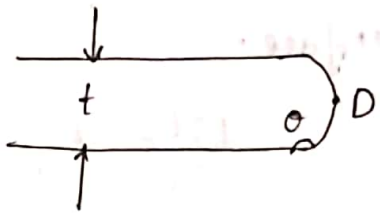
$\Rightarrow \alpha = 8$.

Ex! An amount of water is poured on top of circular face of a horizontal glass cylinder. Excess amount of water spills over the edges and a little amount of water remains their in a thin uniform layer a portion of which is shown in figure. If contact angle is θ and surface tension of water is σ the vertical thickness of water layer far away from the edge is t. The pt in the plane of the diagram on the liquid where the liquid protrudes the most is D. the vertical depth of this pt from the horizontal free surface of water layer is d, s is density of

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water, g is acceleration due to gravity. Mark the correct alternatives.

A) $t = \sqrt{\frac{2\sigma(1+\sin\theta)}{\rho g}}$



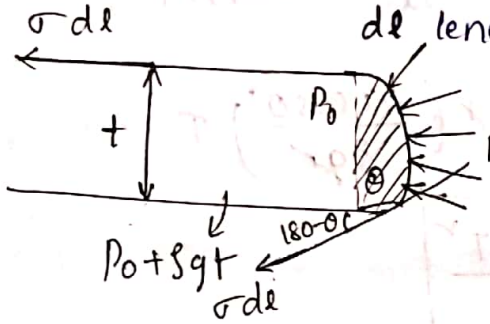
B) $t = \sqrt{\frac{2\sigma(1-\cos\theta)}{\rho g}}$

C) $d = \sqrt{\frac{2\sigma}{\rho g}}$

D) $d = \sqrt{\frac{2\sigma(1-\cos\theta)}{\rho g}}$

Sol^n

①



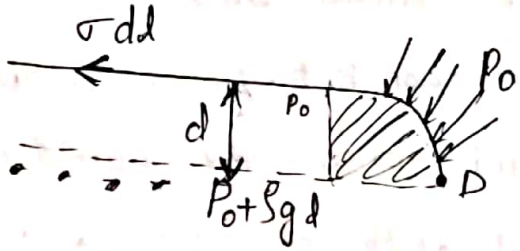
In eqm,

$$\sigma\,dl + \sigma\,dl\cos(180-\theta) + P_0 t\,dl = \left(\frac{P_0 + P_0 + \rho g t}{2}\right) t\,dl$$

$$\sigma(1-\cos\theta) + P_0 t = P_0 t + \frac{\rho g t^2}{2}$$

$$\sqrt{\frac{2\sigma(1-\cos\theta)}{\rho g}} = t$$

②



In eqm,

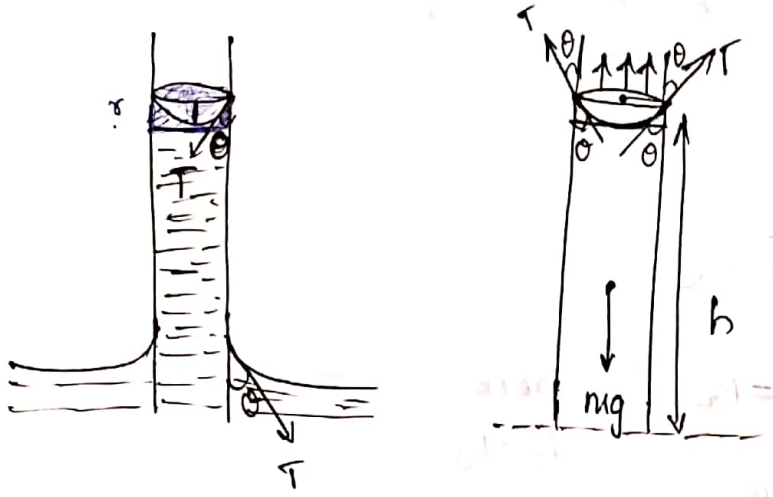
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$$T \cdot dl + P_0 (d) \cdot dl = \left(\frac{P_0 + P_0 + sgd}{2}\right)(d) \cdot dl$$

$$T + P_0 d = P_0 d + \frac{sgd^2}{2}$$

$$\boxed{\sqrt{\frac{2T}{sg}} = d}$$

Surface tension of liquid by cappillary rise method:



* Force required to held the capillary tube in fixed position,

$$F = \text{wt of tube} + 2\pi r T \cos\theta \times 2$$

M-I :

wt balance method :

$$2\pi r T \cos\theta = mg$$

$$2\pi r T \cos\theta = \pi r^2 h s g$$

$$\boxed{h = \frac{2T\cos\theta}{rsg}} *$$

$$\boxed{T = \frac{hrsg}{2\cos\theta}}$$



If the liquid inside the meniscus is also considered

$$2\pi r T \cos\theta = \left(\pi r^2 h + \left[\pi r^2 r - \frac{2}{3}\pi r^3\right]\right) sg$$

$$2T\cos\theta = \left(rh + r^2 - \frac{2}{3}r^2\right) sg$$

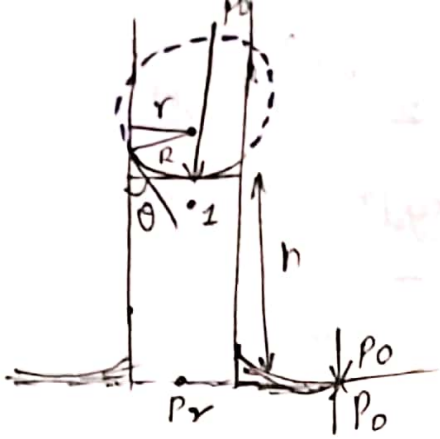
$$2T\cos\theta = \left(rh + \frac{r^2}{3}\right) sg$$

$$\boxed{T = \frac{r\left(h + \frac{r}{3}\right) sg}{2\cos\theta}}$$

M-II :

Pressure balance method :

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$\cos\theta = \frac{r}{R}$ $\Rightarrow$ r = radius of tube

R = radius of meniscus

$\Rightarrow R = \frac{r}{\cos\theta}$

$$P_0 - \frac{2T}{R} = P_1$$

$$P_0 - P_1 = \frac{2T}{R} \quad -(1)$$

and $P_1 + sgh = P_2$ ... here $P_2 = P_0$

$$P_1 + sgh = P_0$$

$$P_0 - P_1 = sgh \quad -(2)$$

from eqns (1), (2)

$$\frac{2T}{R} = sgh$$

$$\frac{2T}{r/\cos\theta} = sgh$$

$$\frac{2T\cos\theta}{r} = sgh$$

$$\boxed{T = \frac{sghr}{2\cos\theta}}$$

Important points:

(1) If the liquid column is in eqm,

$$2\pi rT\cos\theta = mg$$

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tube change — r changes

liquid change — T, S, contact angle changes

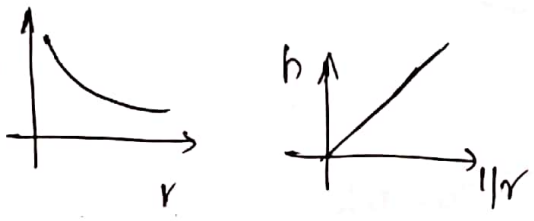
place change — g changes

here $\boxed{m \propto r}$ and $\boxed{V \propto r}$

② Raise of liquid in capillary tube,

$$\boxed{h = \frac{2T\cos\theta}{rsg}}$$

a) $h \propto \frac{1}{r}$



$$h_1 r_1 = h_2 r_2$$

b) * $h \propto \frac{1}{g}$ → experiment is repeated at different places.

$$h_1 g_1 = h_2 g_2$$

value of g is

① at a height h above the earth's surface,

$$\boxed{g_h = g\left(\frac{R}{R+h}\right)^2}$$

② If $h \ll R$ (lower altitudes)

$$g_h = g\left(1+\frac{h}{R}\right)^{-2}$$

$$\boxed{g_h = g\left(1-\frac{2h}{R}\right)}$$

② at a depth d below the earth's surface,

$$\boxed{g_d = g\left(1-\frac{d}{R}\right)}$$

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at the centre of earth, $d = R$

$$\boxed{g_d = 0}^{*}$$

c) In artificial satellites

$$\boxed{g_{eff} = 0}^{*}$$

d) In lift problems

$$\boxed{g_{eff} = g \pm a}$$

* a = acceleration of lift.

$$h_1 g_1 = h_2 g_2$$

$$h_2 = \frac{h_1 g_1}{g_2}$$

If $\boxed{g_2 = 0}^{*}$ then $h_2 \longrightarrow \infty$

③ potential energy stored in a liquid column,

$$U = mg\frac{h}{2}$$



$$U = \pi r^2 h \rho g \frac{h}{2}$$

$$= \frac{\pi r^2 h^2 \rho g}{2}$$

$$= \frac{\pi r^2}{2}\left(\frac{2T\cos\theta}{r\rho g}\right)^2 \rho g$$

$$= \frac{\pi r^2}{2}\frac{(4T^2\cos^2\theta)}{r^2\rho^2 g^2}\cdot \rho g$$

$$\boxed{U = \frac{2\pi T^2\cos^2\theta}{\rho g}}$$

P.E is independent of radius of tube.

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(4) Heat produced;

$$U_i + W = U_f + H$$

work done by surface tension,

$$W_{ST} = (2\pi r T \cos\theta)\, h$$

$\rightarrow PE$

$$0 + (2\pi r T \cos\theta)h = \frac{2\pi T^2 \cos^2\theta}{\rho g} + H$$

$$H = 2\pi r T \cos\theta \cdot \frac{2\pi \cos\theta}{\rho g} - \frac{2\pi T^2 \cos^2\theta}{\rho g}$$

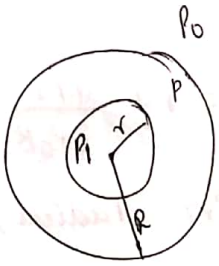
$$H = \frac{4\pi T^2 \cos^2\theta}{\rho g} - \frac{2\pi T^2 \cos^2\theta}{\rho g}$$

$$\boxed{H = \frac{2\pi T^2 \cos^2\theta}{\rho g}}$$

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Ex: A soap bubble of radius 'r' is formed in a soap bubble of radius R. If atm pressure is P_0, surface tension of bubble is T. Calculate change in radius if outer bubble burst. Assume that excess pressure is less than P_0.

Sol:



$$P_0 + \frac{4T}{R} = P$$

and $$P + \frac{4T}{r} = P_1$$

$$\boxed{P_1 = P_0 + \frac{4T}{R} + \frac{4T}{r}}$$

Volume, $V_1 = \frac{4}{3}\pi r^3$

when outer bubble burst,



$$P_0 + \frac{4T}{r_1} = P_1'$$

$$V_1' = \frac{4}{3}\pi r_1^3$$

Boyle's Law,

$$P_1 V_1 = P_1' V_1'$$

$$\left(P_0 + \frac{4T}{r} + \frac{4T}{R}\right) r^3 = \left(P_0 + \frac{4T}{r_1}\right) r_1^3$$

$$\Rightarrow r_1^3 = r^3 \frac{\left(P_0 + \frac{4T}{r} + \frac{4T}{R}\right)}{P_0 + \frac{4T}{r_1}}$$

$$r_1^3 = r^3 \left(\frac{P_0 + \frac{4T}{r}}{P_0 + \frac{4T}{r_1}} + \frac{\frac{4T}{R}}{P_0 + \frac{4T}{r_1}} \right)$$

$$\boxed{r_1 < r}$$

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$$r_1^3 = r^3\left(1+\frac{4T}{P_0R}\right)$$

$$r_1 = r\left(1+\frac{4T}{P_0R}\right)^{1/3}$$

$$r_1 = r\left(1+\frac{4T}{3P_0R}\right)$$

$$r_1 = r+\frac{4Tr}{3P_0R}$$

Increase in radius, $\Delta r = r_1 - r = \frac{4Tr}{3P_0R}$

Ex: A narrow capillary tube open at both ends having radius $r = 0.05\,mm$ contains some water in it. When the tube is held vertical, water remains inside the tube as shown. Given Surface tension $= 0.075\,N/m$, $g = 10\,m/s^2$, $\theta = 0°$. Choose the correct options.



a) for $h = 4\,cm$, lower meniscus is convex upwards.

b) for $h = 6\,cm$, lower meniscus is flat

c) for $h = 8\,cm$, lower meniscus is concave upwards.

d) for any value of h, low meniscus is convex upwards always.



Note:



$\sin(\theta - 90) = \frac{r}{R}$

$-\cos\theta = \frac{r}{R}$

$$P_0 - \frac{2T}{r} + \rho gh = P_2$$

$$P_2 = P_0 - \frac{2\times 0.075}{0.25\times 10^{-3}} + 10^3\times 10\times h$$

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$$\boxed{P_2 = P_0 - 6\times10^{2} + h\times10^{4}}$$

① for $h = 4cm = 4\times10^{-2}\,m$

$$P_2 = P_0 - 6\times10^{2} + 4\times10^{2}$$

$$P_2 = P_0 - 2\times10^{2}$$

$$\Rightarrow \boxed{P_2 < P_0}$$



convex upwards

② for $h = 6cm = 6\times10^{-2}\,m$

$$\boxed{P_2 = P_0}$$

⇒ flat

③ for $h = 8cm = 6\times10^{-2}\,m$

$$P_2 = P_0 + 2\times10^{2}$$

$$\Rightarrow \boxed{P_2 > P_0}$$

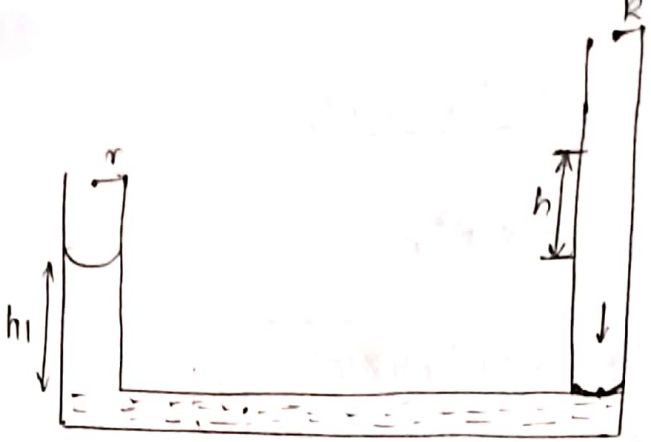
concave upwards

Ex: Left tube of radius r is of short length & right tube of radius R is very long. water is slowly poured in the right tube, $\theta = 60°$. Let h be the height difference b/w water surface in the right & left tube. T is surface tension of water & S is density of water.

1) Find the value of h if the water surface in the left tube is found to be flat.

2) Find the maximum value of h for which water will not flow out of the left tube.

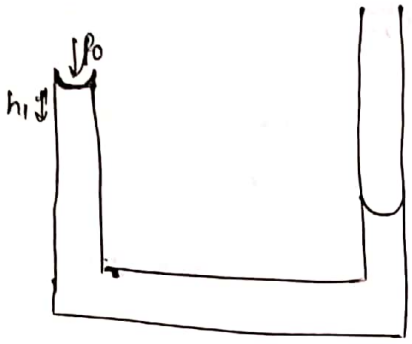
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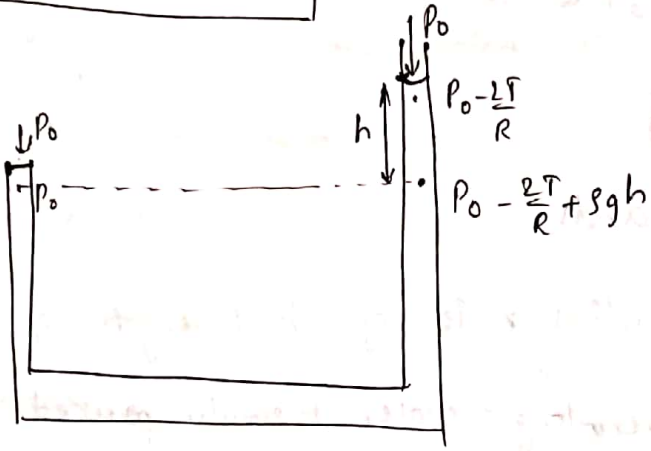
$$P_0 - \frac{2T}{r} + h_1 \rho g = P_0 - \frac{2T}{R}$$

$$h_1 \rho g = 2T\left(\frac{1}{r} - \frac{1}{R}\right)$$

$$h_1 = \frac{2T}{\rho g}\left(\frac{1}{r} - \frac{1}{R}\right)$$



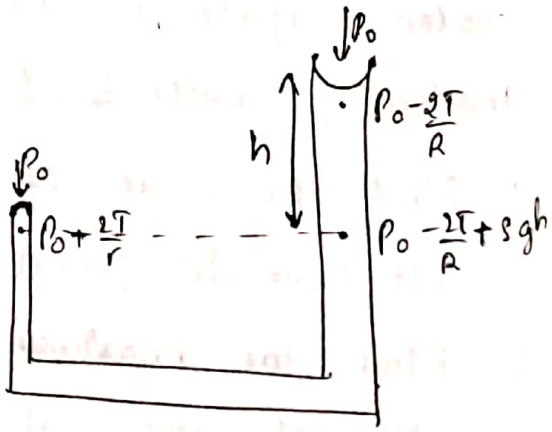
(1)



$$P_0 = P_0 - \frac{2T}{R} + \rho g h$$

$$\boxed{h = \frac{2T}{R \rho g}}$$

(2)



$$P_0 + \frac{2T}{r} = P_0 - \frac{2T}{R} + \rho g h$$

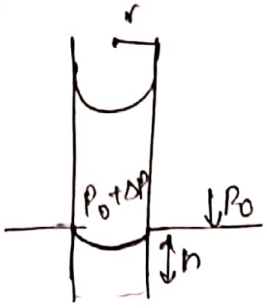
$$\rho g h = \frac{2T}{r} + \frac{2T}{R}$$

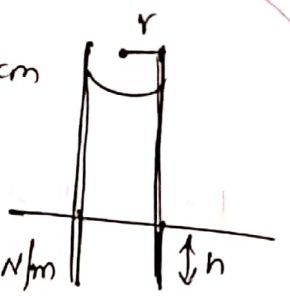
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$$h_{max} = \frac{2T}{\rho g}\left(\frac{1}{r}+\frac{1}{R}\right)$$

ex: Find the excess pressure to be applied on the water so that the water level in the tube becomes same as that in the vessel.

Sol:





$r = 0.025\,cm$

$h = 1\,cm$

$\theta = 0°$

$T = 0.075\,N/m$

$P_0 = 10^5\,N/m^2$

$\rho = 10^3\,kg/m^3$

$$P_0 + \Delta P - \frac{2T}{R} = P_0 + \rho g h$$

$$\Delta P = \frac{2T}{R} + \rho g h$$

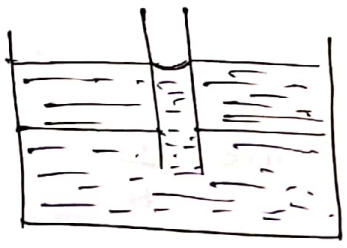
$$= \frac{2 \times 0.075}{0.025 \times 10^{-2}} + 10^3 \times 10 \times 10^{-2}$$

$$\Delta P = 600 + 100 = 700$$

$$P_0 - \frac{2T}{R} + \Delta P = P_0$$

$$\Delta P = \frac{2T}{R} = 600$$

ex:



$$P_0 + \rho_1 g h - \rho_2 g h + \frac{2T}{R} = P_0$$

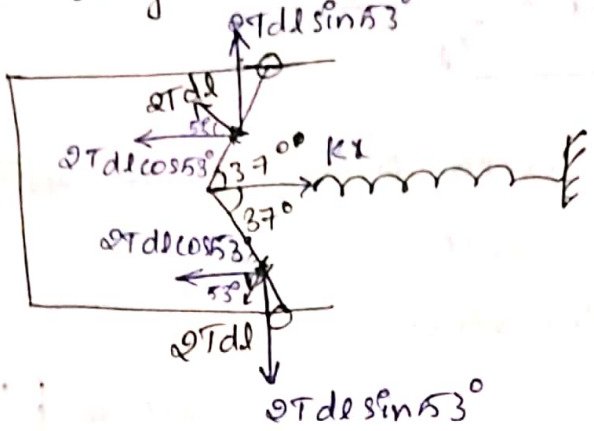
$$\frac{2T}{R} = gh(\rho_2 - \rho_1)$$

$$T = \frac{Rgh}{2}(\rho_2 - \rho_1)$$

ex: A rigid bent rod of total length 2l can slide on fixed wire frame with the help of frictionless sliders. there is thin liquid film (surface tension T) blw bent rod and wire frame.

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In eqm the elongation in spring is l



$$kx = 4Tdl\cos 53^\circ$$

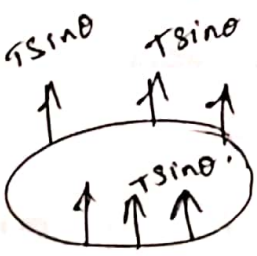
$$kx = 4Tl \times \frac{3}{5}$$

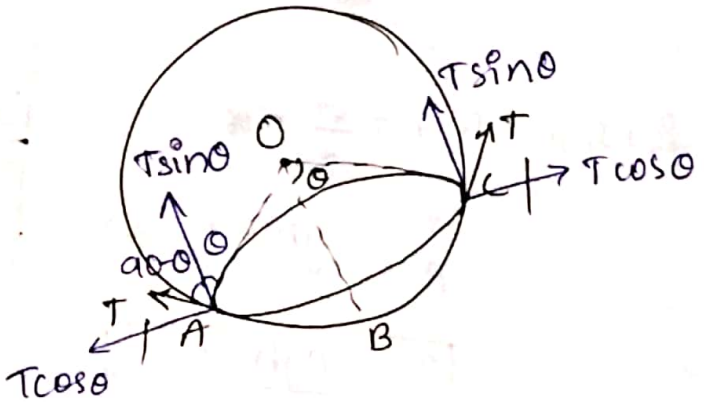
$$\boxed{x = \frac{12Tl}{5K}}$$

Ex: Consider a small water drop in air. let T be the surface tension of water. The fig shows a small section ABC of the drop subtending angle 2θ at the centre O. The force due to surface tension acting on this section is:

Sol:

$$\sin\theta = \frac{r}{R}$$





$$F = T\sin\theta(2\pi r)$$

$$= T\sin\theta(2\pi)(R\sin\theta)$$

$$\boxed{F = 2\pi RT\sin^2\theta}$$

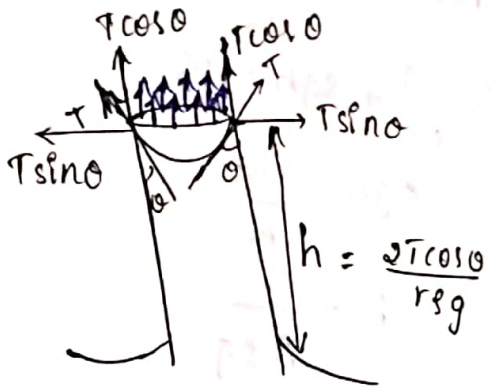
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Work done by surface tension: 30/09/24/ Monday

$W = (2\pi r T\cos\theta)h$

$= 2\pi r T\cos\theta \times \frac{2T\cos\theta}{rsg}$

$$\boxed{W = \frac{4\pi T^2\cos^2\theta}{sg}}$$



From work-energy theorem,

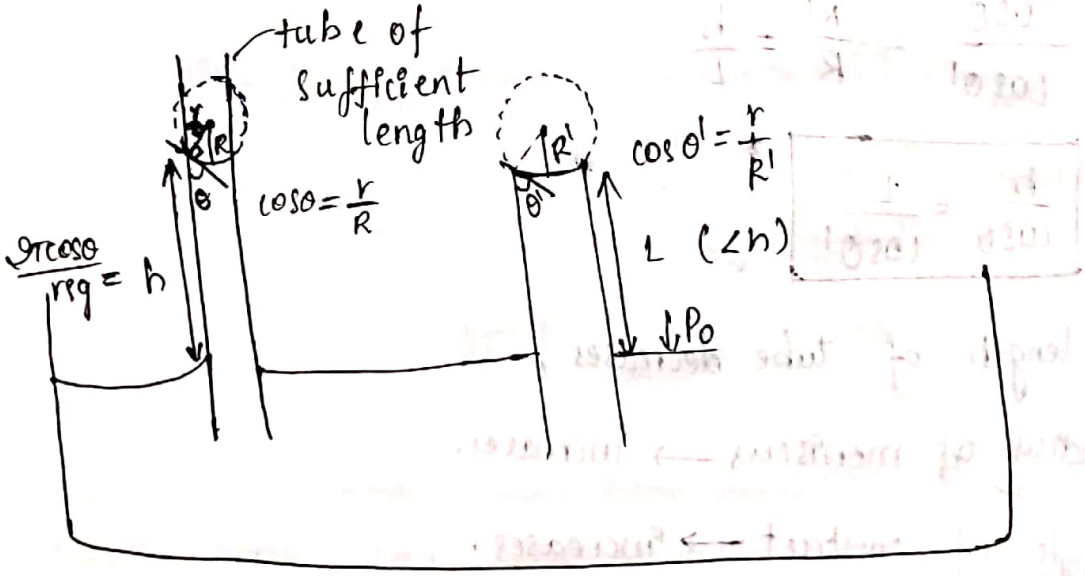
$\cancel{U_i^0} + W = U_f + H$

$H = W - U_f$

$H = \frac{4\pi T^2\cos^2\theta}{sg} - \frac{2\pi T^2\cos^2\theta}{sg}$

$$\boxed{H = \frac{2\pi T^2\cos^2\theta}{sg}}$$

Capillary tube of insufficient length :



* the radius of curvature of meniscus changes but liq will not over flow.

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we have,

$$P_0 - \frac{2T}{R} + \rho g h = P_0$$

$$\rho g h = \frac{2T}{R}$$

$$h = \frac{2T}{R \rho g}$$

$$hR = \frac{2T}{\rho g} \quad -(1)$$

$$P_0 - \frac{2T}{R'} + \rho g L = P_0$$

$$\frac{2T}{R'} = \rho g L$$

$$L = \frac{2T}{R' \rho g}$$

$$LR' = \frac{2T}{\rho g} \quad -(2)$$

(1) = (2) $\Rightarrow$ $\boxed{LR' = hR}$

Again, $\frac{\cos\theta}{\cos\theta'} = \frac{R'}{R} = \frac{h}{L}$

$$\boxed{\frac{h}{\cos\theta} = \frac{L}{\cos\theta'}} \quad *$$ ✓

* As length of tube decreases,

radius of meniscus $\longrightarrow$ increases.

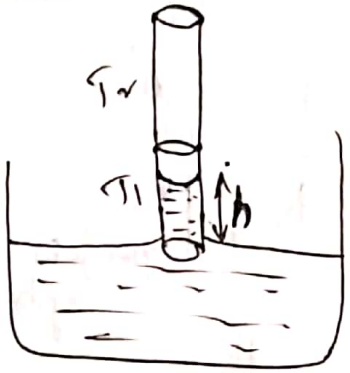
angle of contact $\longrightarrow$ increases.

* As $h > L \Rightarrow \theta' > \theta$.

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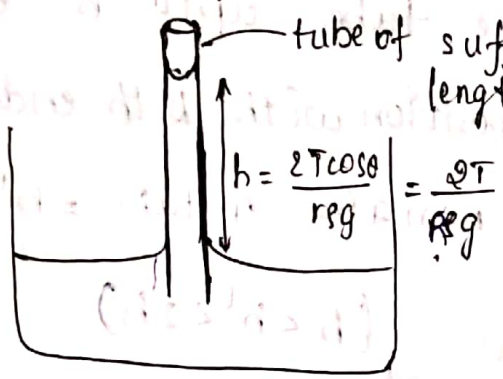
Hybrid Capillary tube:

* Tube made of two materials.
* material changes ⟶ θ changes*.



* Important points:

(1)

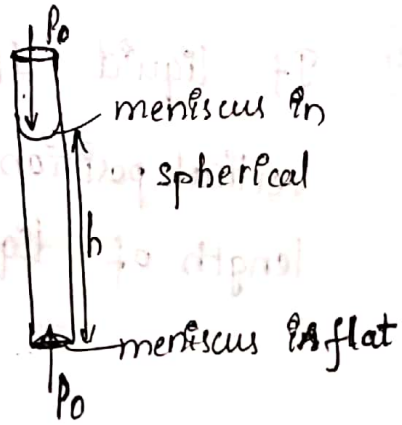


* when the tube is taken out of liquid slowly then h will not change in the tube. when it is completely taken out, length of liquid remains in the tube $= h$.

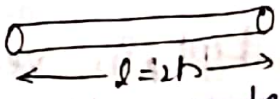
$$P_0 - \frac{2T}{R} + \rho g h = P_0$$

$$\frac{2T}{R} = \rho g h$$

$$\boxed{h = \frac{2T}{R\rho g}}$$

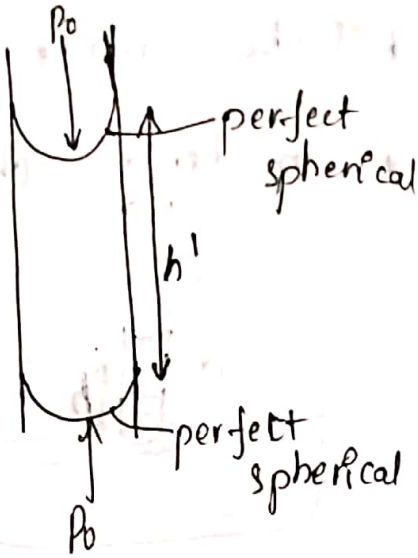


(2) when the same tube is filled with same liquid upto length greater than $2h$ & is held in vertical position with the ends open, then



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length of liquid that remains in capillary tube is $2h$.



$$P_0 - \frac{2T}{R} + \rho g h' - \frac{2T}{R} = P_0$$

$$\rho g h' = \frac{4T}{R}$$

$$h' = \frac{4T}{R\rho g}$$

$$\boxed{h' = 2(h)}$$

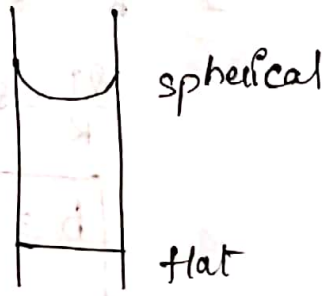
(3) when liquid fills the tube upto h' & tube is kept in vertical position with both ends open. Then length of liquid remains in tube $= h'$



$(h < h' < 2h)$

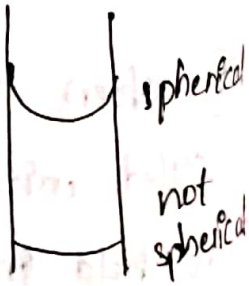
(4) If liquid filled upto height h' & kept in vertical position.

length of liquid in tube $= h$.



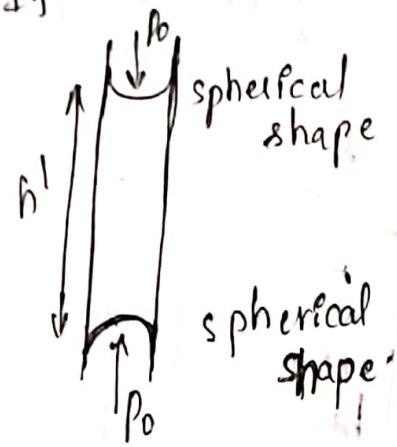
(5) If liquid fills the tube upto $h'(<h)$ and kept in vertical position.

length of liquid $= h'$

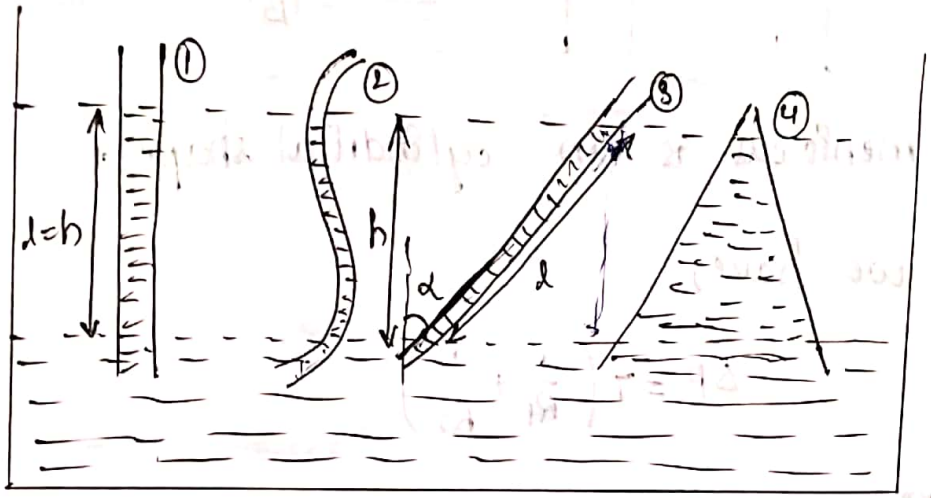


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⑥ If the tube is placed in gravity free region.



☆



Here, ① rise of liquid in all the tubes is same.

tubes

$$\text{i.e, } h = \frac{2T\cos\theta}{\rho g} = \text{same}$$

② length of liquid column in all the tubes will be different.

③ α is the angle of tilt of tube with vertical,

here ① h ② θ] will not change

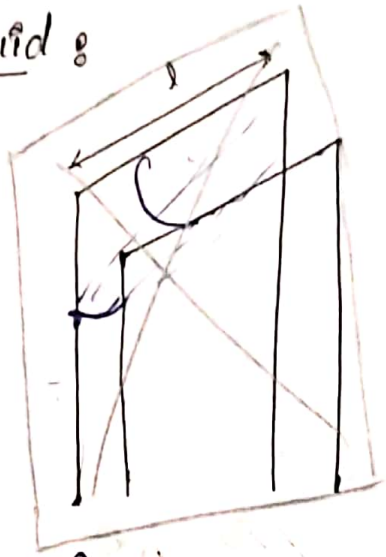
from fig, length of liquid

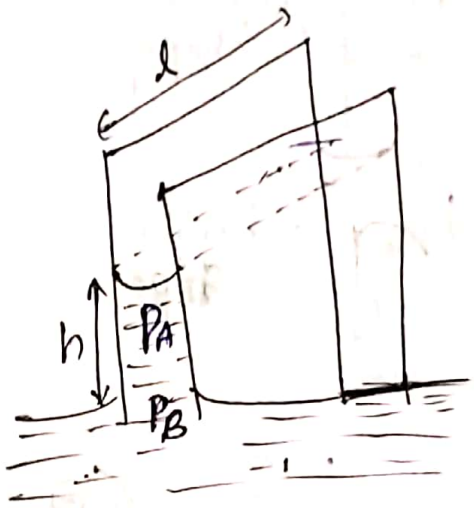
$$\boxed{l = \frac{h}{\cos\alpha}}$$



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Force of attraction b/w the glass plates when dip in liquid:





liquid meniscus is in cylindrical shape.

$\theta = 0°$.

we have,

$$\Delta P = T\left(\frac{1}{R_1} + \frac{1}{R_2}\right)$$

here, $R_1 = d/2$ & $R_2 = \infty$

$$\Delta P = T\left(\frac{2}{d} + \frac{1}{\infty}\right)$$

$$P_0 - P_A = \frac{2T}{d} \quad -①$$

* To find rise of liquid b/w the plates:

weight balance method

$$2Tl = mg$$

$$2T\,l = l\,d\,h\,s\,g$$

$$\boxed{h = \frac{2T}{d\,s\,g}} \quad -②$$

now, force of attraction b/w the plates

$$F = (P_0 - P_{avg}) \times \text{projection area}$$

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$$= \left(P_o - \frac{(P_o + P_A)}{2}\right) lh$$

here $P_o = P_A + \rho g h$

$$F = \left(P_o - \left(\frac{P_A + P_A + \rho g h}{2}\right)\right) lh$$

$$= \left(P_o - P_A - \frac{\rho g h}{2}\right) lh$$

$$F = \left(\frac{2T}{d} - \frac{\rho g h}{2}\right) lh.$$

$$F = \left(\rho g h - \frac{\rho g h}{2}\right) lh$$

$$\boxed{F = \frac{\rho g l h^2}{2}} \quad \text{where } h = \frac{2T}{d\rho g}$$

(or)

$$F = \frac{\rho g l}{2} \times \frac{4T^2}{d^2 \rho^2 g^2}$$

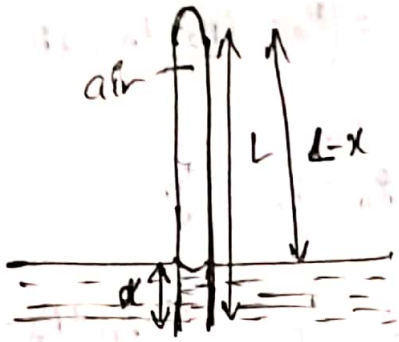
$$\boxed{F = \frac{2T^2 l}{d^2 \rho g}}$$

Ex: A glass capillary sealed at the upper end and is of length 0.11m and internal diameter 2×10^{-5} m. The tube is immersed vertically into a liquid of surface tension 5.06×10^{-2} N/m i) To what length has the capillary to be immersed so that the liq level inside & outside the capillary remains the same ii) what will happen to the ~~the~~ water level inside the capillary if the seal is now broken.

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Sol:

From boyle's law



$$P_1V_1 = P_2V_2$$

$$P_0AL = \left(P_0 + \frac{2T}{r}\right)(L-x)A$$

$$P_0L = P_0L - P_0x + \frac{2T}{r}(L-x)$$

$$P_0x + \frac{2T}{r}x = \frac{2T}{r}L$$

$$\boxed{x = \frac{\frac{2TL}{r}}{P_0 + \frac{2T}{r}}}$$

$$x = \frac{\frac{2 \times 5.06 \times 10^{-2} \times 0.11}{1 \times 10^{-5}}}{10^5 + \frac{2 \times 5.06 \times 10^{-2}}{10^{-5}}}$$

$$= \frac{0.22 \times 5.06 \times 10^{-2}}{1 + 2 \times 5.06 \times 10^{-2}}$$

$$= \frac{0.11 \times 10.12 \times 10^{-2}}{1 + 10.12 \times 10^{-2}}$$

$$= \frac{1.1132 \times 10^{-2}}{1 + 0.1012} = \frac{1.1132 \times 10^{-2}}{1.1012} \approx 0.01$$

$$\boxed{x = 0.01\,m}$$

when the seal is broken now, pressure becomes P_0.

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rise of liquid in the tube,

$$h = \frac{2T}{rsg}$$

$$h = \frac{2 \times 5.06 \times 10^{-2}}{10^{-5} \times 10^{3} \times 4.9}$$

$$h = \frac{5.06}{4.9}$$

$$h = \frac{506}{490} \approx 1.03$$

$$\boxed{h = 1.03m}$$

where as length of tube above liquid

$$= L - x = 0.11 - 0.01 = 0.1m$$

i.e, tube is of insufficient length,

Here liq will not overflow,

$$hr = (l-x)R$$

$$R = \frac{hr}{l-x}$$

$$R = \frac{1.03 \times 10^{-2} \times 4}{0.1}$$

$$\boxed{R = 1.03 \times 10^{-4}}$$

Now,

$$\frac{h}{\cos\theta} = \frac{l-x}{\cos\theta_1}$$

$$\frac{1.03}{\cos\theta} = \frac{0.1}{\cos\theta_1}$$

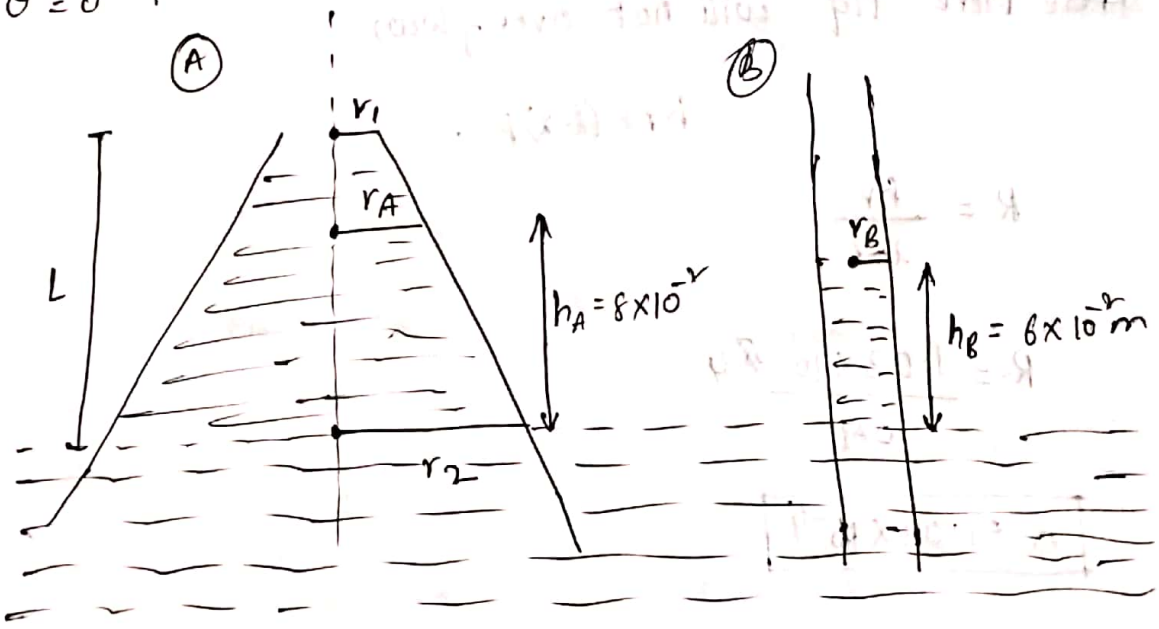
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$$\frac{10\cdot3}{\cos\theta} = \frac{1}{\cos\theta'} \Rightarrow \cos\theta' = \frac{1}{10\cdot3}$$

$$\boxed{\theta' = \cos^{-1}\left(\frac{1}{10\cdot3}\right)}$$

Ex!
A conical glass capillary tube of length 0.1m has diameters 8×10^{-3} m & 5×10^{-4} m at the ends. when it is just immersed in a liquid at zero °c with larger diameter in contact with it, the liquid rises to 8×10^{-2} m in the tube. If another cylindrical glass capillary tube B is immersed in the same liquid at zero °c the liquid rises to 6×10^{-2} m height. The rise of liquid in tube B is only 5.5×10^{-2} m when the liquid is at 50°c. Find the rate at which surface tension changes with temperature. considering the change to be linear. $\rho_{liq} = \frac{1}{14}\times10^{4}$ kg/m³

$\theta = 0°$.



* If $r_A = r_B$ then $h_A = h_B$. But here $h_A \neq h_B$. So $r_A \neq r_B$

for tube A

$$r_A = r_1 - \left(\frac{r_1 - r_2}{L}\right) h_A$$

$L \longrightarrow r_1 - r_2$

$1 \longrightarrow \frac{r_1 - r_2}{L}$

$h_A \longrightarrow \left(\frac{r_1 - r_2}{L}\right) h_A$

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$$2r_A = 2r_1 - \left(\frac{2r_1 - 2r_2}{L}\right) h_A$$

$$2r_A = 10^{-3} - \left(\frac{10^{-3} - 0.5 \times 10^{-4}}{001}\right) 8 \times 10^{-2}$$

$$\boxed{r_A = 3 \times 10^{-4} m}$$

$$T = \frac{h r \rho g}{2}$$

$$h_B r_B = h_A r_A \quad \text{at } 0^\circ C$$

$$6 \times 10^{-2} \times r_B = 3 \times 10^{-4} \times 8 \times 10^{-2}$$

$$\boxed{r_B = 4 \times 10^{-4}}$$

now, for tube B

$$T = \frac{h_0 r \rho g}{2}$$

$$= \frac{6 \times 10^{-2} \times 14 \times 10^{-4} \times \frac{1}{14} \times 10^{4} \times 10}{2} = \frac{60}{7} \times 10^{-2}$$

$$\boxed{T = 8.4 \times 10^{-2} N/m}$$

we have,

$$T \propto h$$

$$\frac{T_0}{h_0} = \frac{T_{50}}{h_{50}} \Rightarrow \frac{8.4 \times 10^{-2}}{6 \times 10^{-2}} = \frac{T_{50}}{5.5 \times 10^{-2}}$$

$$\Rightarrow \boxed{T_{50} = 7.7 \times 10^{-2} N/m}$$

finally,

$$\frac{\Delta T}{\Delta \theta} = \frac{T_{50} - T_0}{50 - 0} = \frac{7.7 \times 10^{-2} - 8.4 \times 10^{-2}}{50}$$

$$\boxed{\frac{\Delta T}{\Delta \theta} = -1.4 \times 10^{-4}}$$

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-ve sign indicates decrease in surface tension

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